

Lebanon Healthcare & Population Impact Dashboard Visualization Analysis Report

DSC 614 - Data Visualization

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1 Introduction

1.1 Domain Situation

The Lebanon Healthcare & Population Impact Dashboard targets three primary user groups: public health analysts and policymakers who must monitor healthcare resource allocation in a crisis context; academic researchers comparing Lebanon's health indicators with regional and European peers; and informed citizens or journalists seeking to understand the lived impact of Lebanon's economic collapse on healthcare affordability.

Lebanon's healthcare system operates within one of the most turbulent economic environments in recent history. The country's financial collapse, which accelerated after the 2019 uprising and the August 2020 Beirut port explosion, produced hyperinflation in medical goods, a dramatic devaluation of the Lebanese pound, mass emigration of healthcare workers, and a steep decline in government health funding. Understanding how these forces interact with population dynamics (fertility, migration, aging) requires more than summary statistics; it demands a visual system that allows users to explore multiple indicators across time and across comparable countries simultaneously.

1.2 Motivation for Visualization

Automated statistical summaries alone are insufficient here for three reasons rooted in Munzner's justification criteria. First, the dataset's multivariate structure, covering 15 years, 5 countries, and 20+ health and demographic indicators, cannot be reduced to a single meaningful metric. Second, the analytical questions are partially open-ended: users do not always know in advance which indicators will show the most alarming trends, making exploratory visualization essential. Third, the temporal dynamics are non-linear and crisis-driven; the Medical CPI for Lebanon, for example, spans more than five orders of magnitude, a pattern that pure tabular summaries would render incomprehensible.

1.3 Goals of the Dashboard

The dashboard is designed to answer four high-level analytical questions:

1. How has healthcare expenditure and medical inflation evolved in Lebanon between 2010 and 2024, and how does the post-2019 crisis appear in the data?
2. How does Lebanon compare with Egypt, France, Syria, and Türkiye on key health and economic indicators?
3. What is the geographic distribution of hospital infrastructure across Lebanon's governorates?

4. How do economic capacity (GDP per capita) and health system burden (out-of-pocket expenditure) relate across countries?

1.4 Report Structure

This report covers each of the six tabs of the dashboard: Regional Overview, Lebanon in Crisis, Healthcare Expenditure, Demographics, Economy & Affordability, and Indicator Comparison, with a per-tab analysis of all KPI cards, charts, and interactive elements.

For every chart the analysis follows the *What / Why / How* framework, supplemented by assessments of expressiveness, effectiveness, color mapping, and scalability. Global interaction mechanisms (the year slider and country filter) are described in Section 5.

2 Dataset Description & Data Abstraction

2.1 Data Sources and Integration

The dashboard integrates five distinct datasets merged into a single flat table keyed by $\text{Country} \times \text{Year}$:

1. **United Nations Population Division - World Population Prospects:** total population, gender-disaggregated population, birth rate, death rate, fertility rate, life expectancy (overall, male, female), infant mortality, net migration rate, and median age (2010–2024).
2. **World Bank Open Data:** GDP (current USD), GDP per capita, health expenditure as % of GDP, health expenditure per capita (USD), government health expenditure as % of government spending, and out-of-pocket health expenditure as % of total health expenditure (2010-2023; 2024 values interpolated or carried forward).
3. **Central Administration of Statistics (CAS) Lebanon:** Consumer Price Index, Health Division (base December 2013 = 100).
4. **Ministry of Public Health Lebanon:** count of hospitals by governorate.
5. **Dashboard-computed derived attributes:** normalized radar scores (0-100), log-transformed CPI, and year-over-year delta fields.

2.2 Attribute Types and Semantics

Following Munzner’s data type taxonomy, each attribute is classified as categorical (supports equality comparison only), ordinal (supports ordering), or quantitative (supports

arithmetic). The direction column distinguishes sequential attributes (only positive magnitudes) from diverging attributes (meaningful zero crossing) and is used to guide colormap selection.

Attribute	Type	Direction	Role	Notes
Country	Categorical	–	Key	5 categories
Year	Ordinal	Sequential	Key	2010–2024
Population (M)	Quantitative	Sequential	Value	Millions
Median Age (yrs)	Quantitative	Sequential	Value	
Pop Growth Rate	Quantitative	Diverging	Value	Can be negative
Birth Rate	Quantitative	Sequential	Value	Per 1,000
Fertility Rate	Quantitative	Sequential	Value	Children per woman
Death Rate	Quantitative	Sequential	Value	Per 1,000
Life Expectancy	Quantitative	Sequential	Value	Years
Infant Mortality	Quantitative	Sequential	Value	Per 1,000 live births
Net Migration Rate	Quantitative	Diverging	Value	Can be negative
GDP per Capita	Quantitative	Sequential	Value	Current USD
Health Exp % GDP	Quantitative	Sequential	Value	Percentage
Health Exp/Capita	Quantitative	Sequential	Value	USD
Govt Health % Exp	Quantitative	Sequential	Value	Percentage
Out-of-Pocket %	Quantitative	Sequential	Value	Percentage
Lebanon CPI (YoY)	Quantitative	Sequential	Value	% change
Governorate	Categorical	–	Key	6 regions
Hospital Count	Quantitative	Sequential	Value	Integer count

The two diverging attributes (Pop Growth Rate and Net Migration Rate) use a zero-baseline bar chart so that the zero reference line is always visible, which is the appropriate encoding for diverging data. A diverging colormap was not applied to these attributes in heatmap views because they do not appear as heatmap columns; the sequential blue colormap used elsewhere is therefore not a mismatch.

3 Task Abstraction

Following Munzner’s {action, target} pair vocabulary, tasks are decomposed into three levels: analyze (high-level intent), search (how the target is found), and query (what is

asked about the target once found).

The dashboard supports three high-level analyze actions. **Discover (primary)**: time-series charts and scatter plots surface unexpected findings such as Lebanon’s medical CPI increasing by a factor of roughly 500 between 2018 and 2024. **Present (secondary)**: the hospital distribution bar chart and KPI cards communicate known facts clearly to a non-expert audience. **Derive (supporting)**: the radar chart uses normalized 0-100 scores derived from raw values.

Chart	Search Type	Query Type
Medical CPI Bar (YoY %)	Locate	Identify
Lebanon GDP vs Peers	Browse	Compare
Health Exp Per Capita	Browse	Compare
Hospital Distribution	Lookup	Identify
Demographic Heatmap	Browse	Summarize
GDP vs Health Exp Scatter	Explore	Compare
Govt Health Spending Slope	Browse	Compare
OOP vs GDP Bars	Explore	Compare
Income & Life Exp Scatter	Explore	Compare
GDP Bar / National Income	Lookup	Compare
Country Profile Radar	Browse	Compare
Indicator Comparison Heatmap	Browse	Summarize

Lookup tasks (known target, known location) are supported by KPI cards and the hospital bar chart. Locate tasks (known target, unknown location) are supported by annotated bar charts with dashed reference lines. Browse tasks (unknown target, unknown location) drive the exploratory multi-country line and scatter charts. Explore tasks (unknown target, unknown location) are supported by the scatter plots and the interactive heatmap in Tab 6.

4 Four Nested Levels of Design

Level 1: Domain Situation. Target users include health economists, public policy researchers, and journalists. They require visual aggregation to perceive trends across 15 years and 5 countries simultaneously. The main validity threat is audience mismatch: if

a general user cannot decode axis labels such as “Health % GDP”, the presentation task fails. The design mitigates this with subtitle annotations and tooltip explanations.

Level 2: Data/Task Abstraction. Raw World Bank and UN data was preprocessed to fill missing years, normalize CPI to a common base, and compute per-capita health expenditure. Derived attributes include year-over-year CPI % change and radar normalization scores. The main validity threat is data transformation error.

Level 3: Visual Encoding/Interaction Idiom. The dashboard uses line charts, bar charts, scatter plots, a radar chart, a heatmap matrix, a slope chart, and stacked area charts. Interaction idioms include a year slider, country toggle buttons, metric dropdowns, and tooltips. The main validity threat is overplotting in scatter plots when five country dots cluster near the origin.

Level 4: Algorithm. Data is pre-aggregated at the country-year level. Normalization for the radar chart is computed once at load time. The main validity threat is log₁₀ scale misinterpretation.

5 Dashboard Architecture & Global Controls

5.1 Tab Structure

The dashboard is organized into **six tabs**, each targeting a distinct analytical theme:

Tab	Focus
Regional Overview	Macro comparison across all five countries
Lebanon in Crisis	Post-2019 crisis indicators for Lebanon
Healthcare Expenditure	Spending breakdown and hospital infrastructure
Demographics	Population trends, fertility, aging, mortality
Economy & Affordability	GDP, health burden, government prioritization
Indicator Comparison	Flexible multi-country, multi-year heatmap

The tabbed layout follows juxtaposed multi-view strategy: charts are not superimposed but organized thematically, reducing cognitive overload while allowing users to build a mental model of each topic before moving to the next.

5.2 Year Slider



Figure 1: Global year slider (2010-2024) and country toggle buttons. The selected year updates all KPI cards and point-in-time charts simultaneously.

What. The year slider is a discrete ordinal filter mapping to the range 2010-2024 (15 steps). The numeric box to the right shows the step index (step 14 = 2024).

Why. The slider implements Filter interaction category: it reduces the data space to a single time slice for all point-in-time charts while leaving time-series charts unaffected.

How. Implemented as a Dash `dcc.Slider` component. Callbacks fire on value change and update all figures that depend on a selected-year parameter.

Dynamic KPI behavior: When the slider moves from 2024 to 2019, all four KPI cards on every tab re-render to show Lebanon's values for 2019. The Medical CPI jumps from 42,000 to a much lower value, making the crisis transition immediately visible within a single slider drag.

5.3 Country Toggle Buttons

What. Five pill-shaped country buttons: Lebanon (blue), Egypt (orange), France (green), Syria (purple), and Turkiye (red), allow users to dynamically switch the country displayed in KPI cards and selected visual components. Each button acts as an interactive categorical filter.

Why. The buttons implement filter interaction idiom on the country dimension. Instead of overwhelming the user with all country values simultaneously inside KPI cards, the interaction allows users to focus on one country at a time. This reduces cognitive overload and supports rapid comparison between national profiles.

How. Clicking a country button updates the KPI cards instantly while preserving the current year selected in the global slider. Active buttons appear filled with their assigned country color, while inactive buttons remain outlined. The interaction is implemented using Dash callbacks connected to the selected-country state variable.

5.3.1 Country Button Interaction - KPI Switching Example



(a) France selected. KPI cards display affordability, OOP burden, and government share values for France in 2024.

(b) Lebanon selected. KPI cards immediately update to Lebanon's values while maintaining the same selected year.

Figure 2: Country toggle buttons dynamically update KPI cards according to the selected country.

Visualization impact. The country toggle interaction allows users to compare national profiles without changing dashboard structure or chart layout. Since only the KPI values update, users maintain spatial consistency while rapidly switching between countries. This improves readability and supports direct country-to-country comparison.

6 Tab 1: Regional Overview

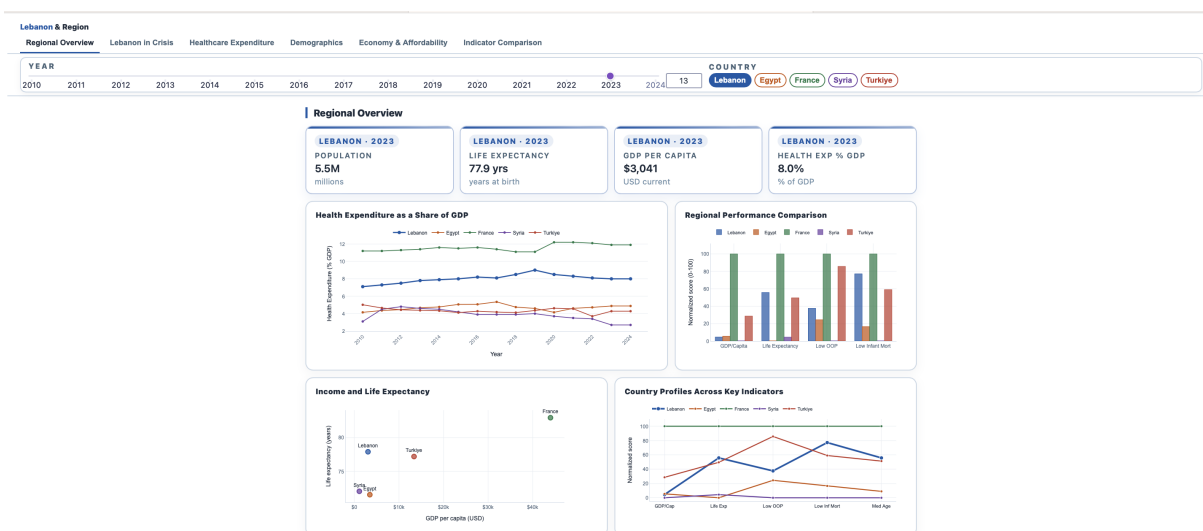


Figure 3: Regional Overview tab - full view: KPI cards, Health Expenditure as a Share of GDP line chart, and Regional Performance Comparison radar chart (2024).

6.1 KPI Cards

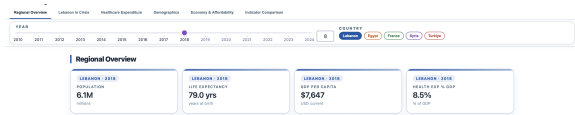
Card	Value (2024)	Unit	Role
Population	5.3M	millions	Overview
Life Expectancy	77.1 yrs	years at birth	Health outcome
GDP per Capita	\$3,150	USD current	Economic context
Health Exp % GDP	8.0%	% of GDP	Health spending share

What. Each card encodes a single quantitative value for Lebanon at the selected year.

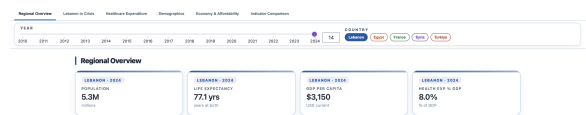
Why. Present / Lookup / Identify pattern: the cards are summary anchors that orient the user before exploring the charts. They eliminate the need to extract headline numbers from denser chart encodings.

How. Large bold numerals with label subtitles. A blue-tinted card background (#1558A8 tint) and a “LEBANON · 2024” tag provide explicit provenance. As the year slider changes, values re-render; the GDP per Capita card drops dramatically from \$8,200 (2018) to \$3,150 (2024).

Country-sensitivity. The KPI cards always display Lebanon’s values regardless of which country toggles are active. Lebanon is the focal country; the cards provide a stable anchor even while charts below show all five countries.



(a) KPI cards at year 2018 (pre-crisis). GDP per Capita approximately \$8,200.



(b) KPI cards at year 2024 (post-crisis). GDP per Capita = \$3,150.

Figure 4: KPI card values change dynamically with the year slider.

6.2 Chart 1: Health Expenditure as a Share of GDP

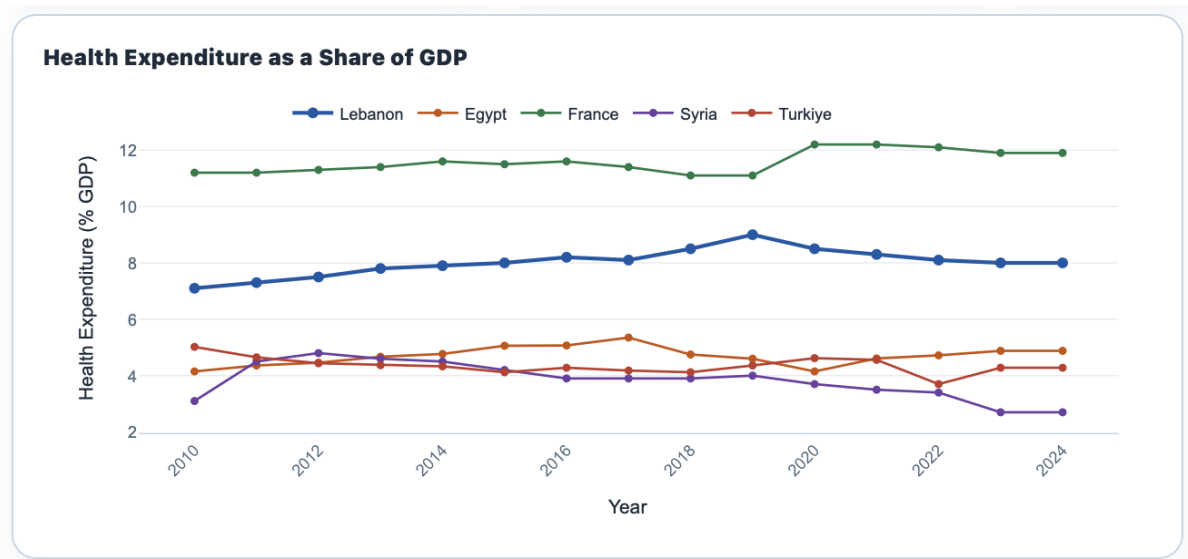


Figure 5: Health Expenditure as a Share of GDP (% of GDP), all five countries, 2010–2024. France (teal) consistently leads near 11–12%; Lebanon (blue) fluctuates between 7 and 9%.

What. Quantitative sequential attribute, Health Expenditure as % of GDP, over ordinal key Year for all five countries. Five series, 15 time points each.

Why. *Discover / Browse / Compare:* track each country's health spending commitment over time. Primary target: *trend*. Secondary target: *extremes* (France as high outlier, Syria declining).

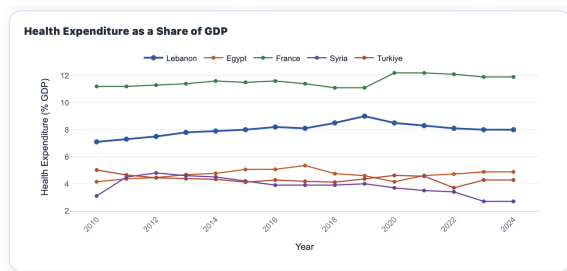
How - Marks and Channels. *Marks:* lines with circular point markers. *Channels:* x = Year (ordinal); y = % GDP (quantitative, aligned, highest-ranked magnitude channel); hue = country (categorical); marker shape = country (redundant identity channel for colorblind users).

Expressiveness. All quantitative series use aligned vertical position; ordinal Year is on the horizontal axis. Country identity uses hue with redundant marker shape.

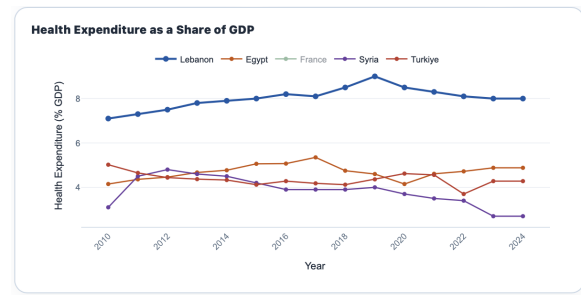
Effectiveness. France's line at 11-12% creates a strong visual separation from the cluster (3-9%), making the gap in health investment immediately apparent.

Alternative encodings rejected. A stacked area chart was rejected because it would imply that the five countries' health spending shares sum to a meaningful total, which they do not. A grouped bar chart for each year was rejected because it would obscure temporal continuity, which is the primary analytical target.

Legend Toggle Effect. Clicking France in the legend removes its dominant high line, rescaling the y-axis and revealing more variation among the remaining four countries.



(a) All five countries. France dominates the upper region.



(b) France removed. Y-axis rescales; differences among remaining four countries become clearer.

Figure 6: Health % GDP line chart — France legend toggle effect.

6.3 Chart 2: Regional Performance Comparison (Radar)

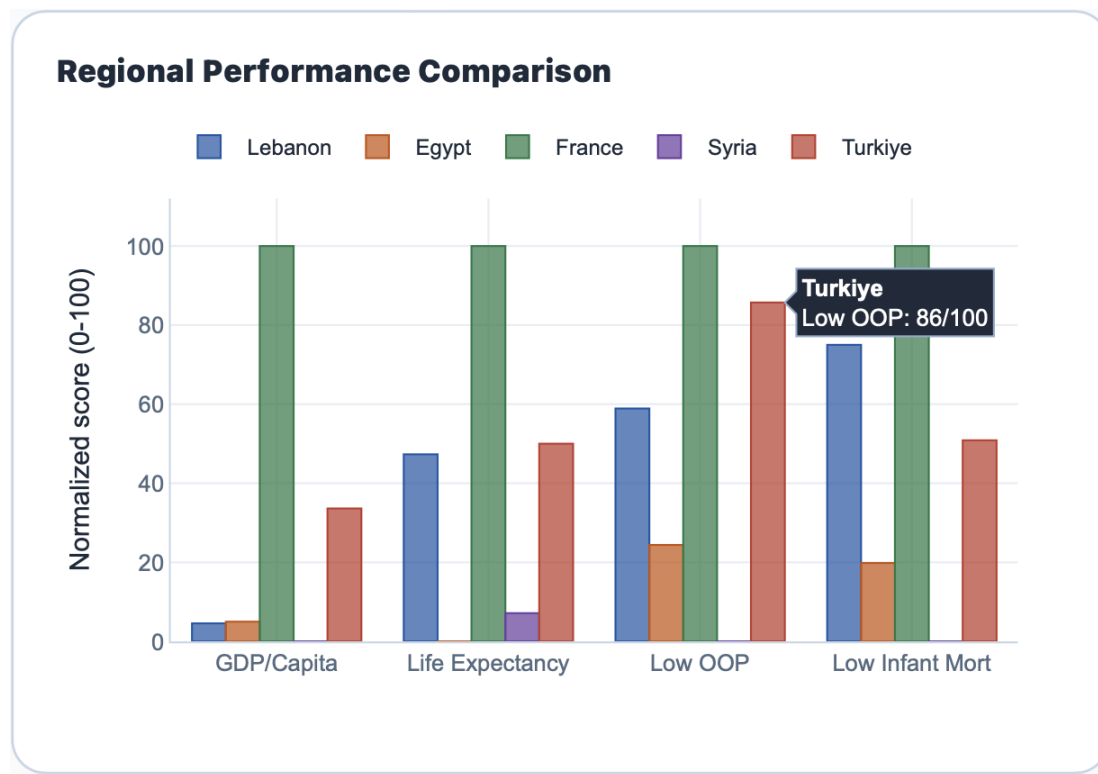


Figure 7: Regional Performance Comparison radar: normalized scores (0-100) for five indicators across all countries at the selected year.

What. Five derived quantitative attributes (normalized 0-100 scores for GDP/Cap, Life Expectancy, Low OOP, Low Infant Mortality, Median Age) for 5 countries at the selected year.

Why. *Browse / Compare / Summarize:* provide a holistic multi-indicator fingerprint before users drill into thematic tabs.

How - Marks and Channels. *Marks:* polygon outlines and semi-transparent filled

areas. *Channels*: radial distance = normalized score (magnitude); angular position = indicator (categorical); hue = country; filled area opacity (redundant identity).

Expressiveness. Radial distance is a lower-ranked magnitude channel than aligned position. However, the five indicators cannot share a single aligned axis; the radar's radial layout is an appropriate multi-attribute idiom that sacrifices individual precision for holistic shape comparison.

Effectiveness. France's polygon is largest and most regular. Lebanon's is mid-size but skewed toward Life Expectancy and away from GDP/Cap. Syria's polygon is narrow, indicating low economic scores.

Alternative encodings rejected. A small-multiples bar chart per indicator was considered, but it would require 25 bars and prevent the holistic country-shape comparison that is the primary task. The parallel coordinates chart (Chart 4) serves the precision task; the radar serves the overview task.

6.4 Chart 3: Income and Life Expectancy (Scatter)

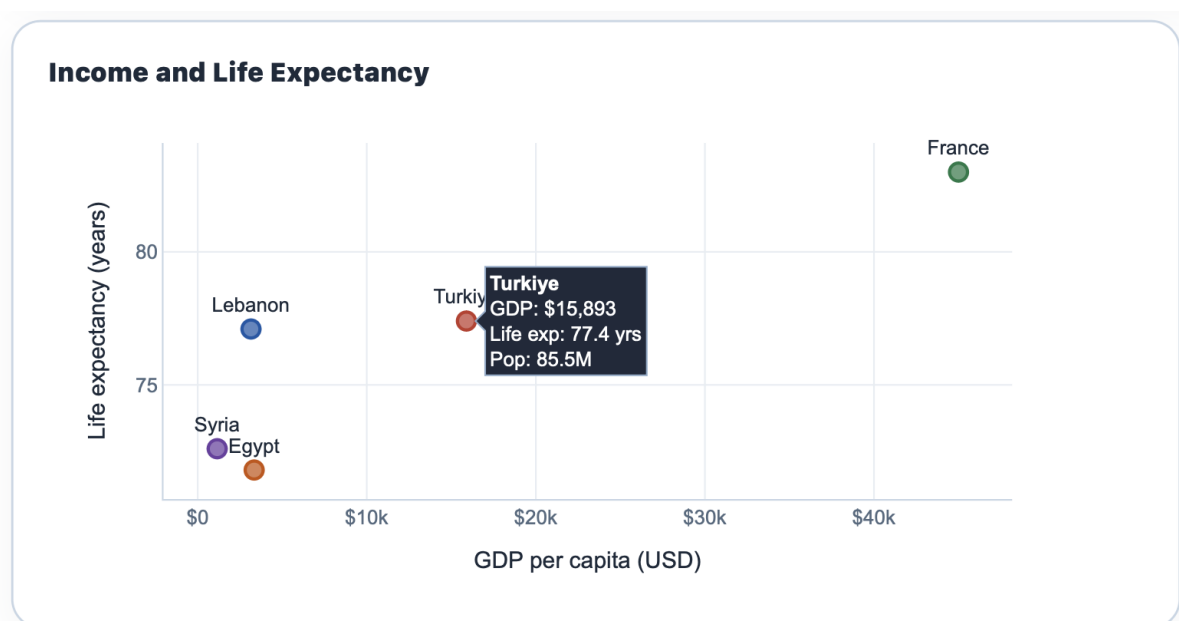


Figure 8: Income and Life Expectancy scatter: GDP per capita (x) vs life expectancy in years (y), one point per country at the selected year. Country labels appear adjacent to each point.

What. Two quantitative sequential attributes: GDP per capita (USD) and Life Expectancy (years), for 5 countries at one selected year.

Why. *Explore / Compare*: discover whether income correlates with longevity. Targets: *correlation* and *outliers* (Lebanon's high life expectancy relative to modest income).

How - Marks and Channels. *Marks*: filled circles with text labels. *Channels*: x = GDP/cap (aligned position); y = Life Expectancy (aligned position); hue = country; text label = country name (redundant identity).

Effectiveness. France occupies top-right (high GDP, high life expectancy). Lebanon and Turkiye cluster mid-range, Lebanon's life expectancy (77.1 yrs) is comparable to Turkiye's despite substantially lower GDP. **Year-Slider Effect.** Moving from 2010 to 2024 shifts Lebanon's point leftward (falling GDP), while y-position changes modestly, tracing the decoupling of economic collapse from immediate mortality outcomes.

Alternative encodings rejected. Two separate bar charts (one per attribute) were rejected because they would require the user to hold one chart in memory while reading the other, violating the eyes-beat-memory principle. The scatter plot encodes the correlation relationship directly using the two highest-ranked channels (x and y position), which is the optimal encoding for two quantitative attributes.

6.5 Chart 4: Country Profiles Across Key Indicators

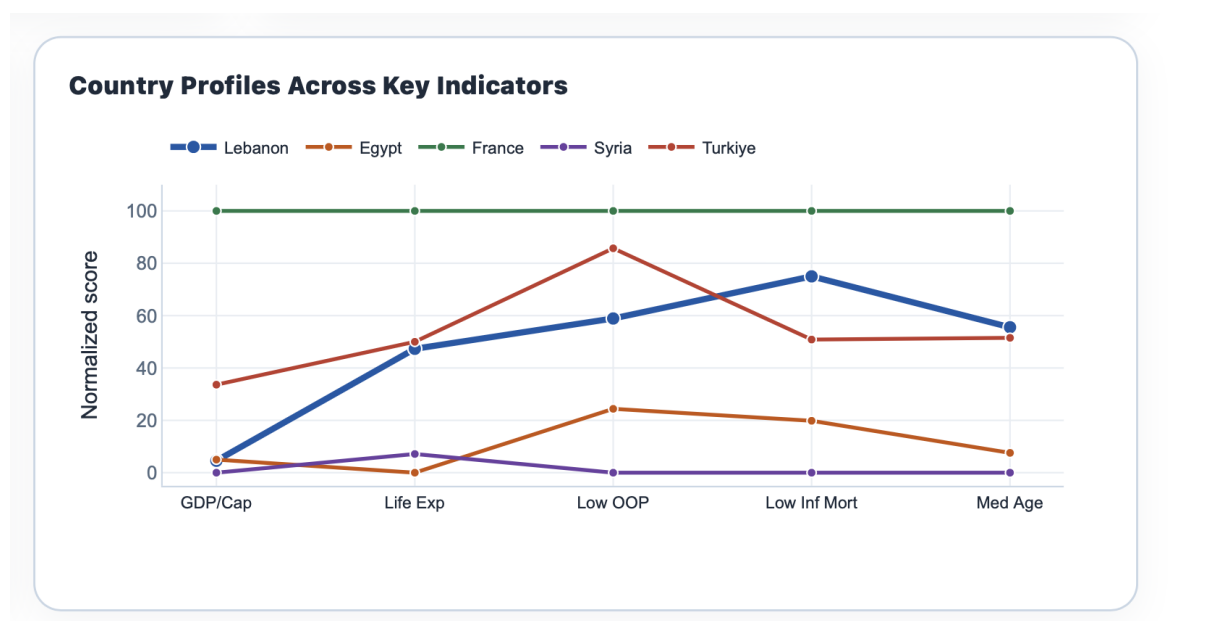


Figure 9: Country Profiles Across Key Indicators: normalized scores for GDP/Cap, Life Exp, Low OOP, Low Infant Mortality, and Median Age as a parallel-coordinates-style line chart.

What. Normalized 0-100 scores on five categorical indicators for 5 countries as connected line segments across parallel categorical x-axis positions.

Why. *Browse / Compare*: identify which countries are consistently strong or weak. Complements the radar by using a Cartesian layout more precise for reading exact per-indicator differences.

How - Marks and Channels. *Marks*: points and polylines. *Channels*: x = indicator (categorical, spatial region); y = normalized score (quantitative, aligned position); hue = country. Unlike the radar, aligned vertical position is used for score magnitude, a higher-ranked channel.

Alternative encodings rejected. A heatmap was considered but would use color luminance for magnitude, a lower-ranked channel than aligned position. This chart and the radar complement each other: the radar supports gestalt shape comparison while this chart supports precise per-axis reading.

7 Tab 2: Lebanon in Crisis

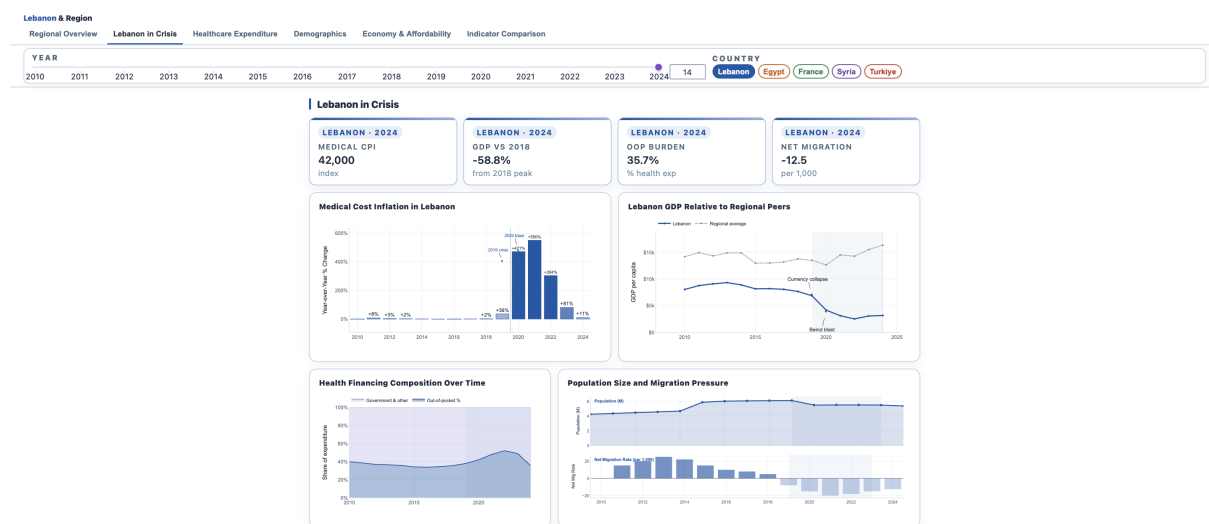


Figure 10: Lebanon in Crisis tab - full view: crisis KPI cards, Medical Cost Inflation bar chart, and Lebanon GDP Relative to Regional Peers.

7.1 KPI Cards

Card	Value (2024)	Unit	Interpretation
Medical CPI	42,000	index	Base Dec 2013 = 100
GDP vs 2018	-58.8%	% change	Relative to 2018 peak
OOP Burden	35.7%	% health exp	Out-of-pocket share
Net Migration	-12.5	per 1,000	Net outflow

What / Why / How. These four cards are crisis indicators, each encodes a scalar derived attribute for Lebanon at the selected year. They function as a shock summary: medical goods cost 420 times their 2013 base, the economy contracted by nearly 60% from its 2018 peak, and 12.5 people per 1,000 are leaving the country. The negative signs in the

large numeral display express the diverging semantics of GDP change and net migration directly.

7.2 Chart 5: Medical Cost Inflation in Lebanon (Bar Chart)

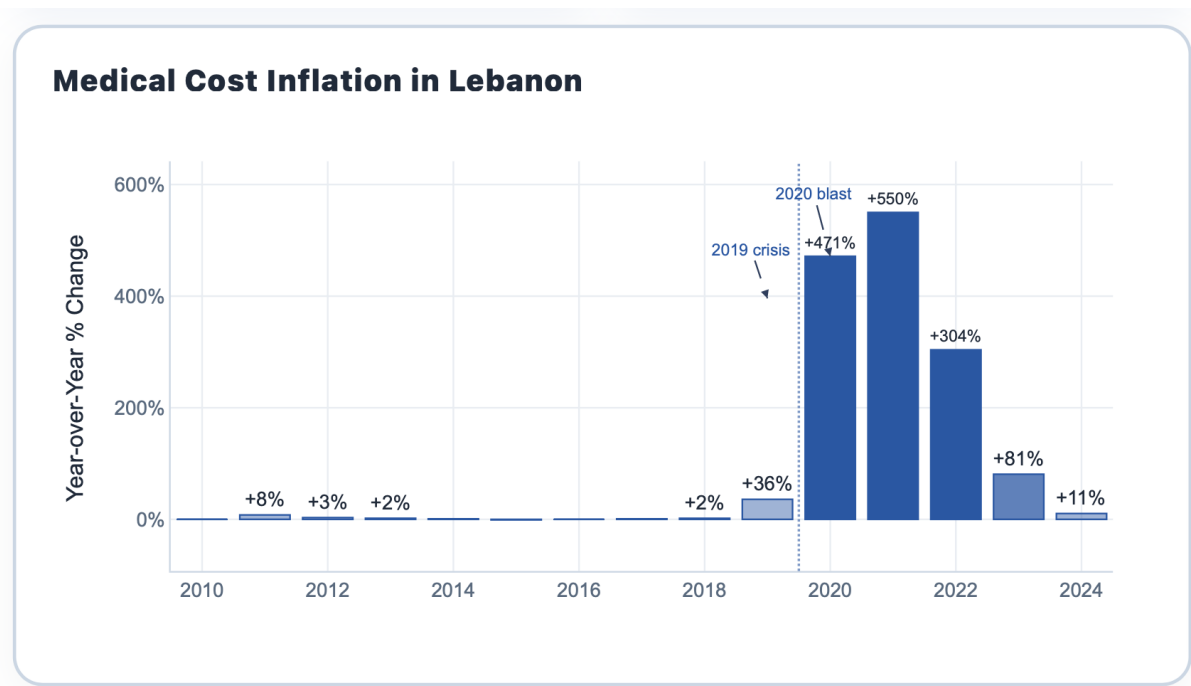


Figure 11: Medical Cost Inflation in Lebanon: year-over-year % change in the CPI Health Division. Annotations show annual rates; dashed lines mark the 2019 crisis and 2020 Beirut blast.

What. Quantitative sequential attribute (YoY % change in Lebanon’s Medical CPI) over Year (2010–2024). Lebanon-only.

Why. *Discover / Locate / Identify:* detect the post-2019 hyperinflationary shock and distinguish it from gradual pre-crisis inflation. Target: *outlier* and *extreme*.

How - Marks and Channels. *Marks:* vertical bars. *Channels:* x = Year (ordinal); bar height = YoY % change (aligned length); text annotations atop bars (e.g., “+8%”, “+500%”, “+304%”); vertical dashed lines at “2019 crisis” and “2020 blast”.

Expressiveness. Bar length from a zero baseline is the correct channel for a ratio-change attribute. Text labels add a redundant numeric channel for precision.

Effectiveness. The 2020-2022 bars are dramatically taller than 2010-2018 bars, making the crisis unmistakable. Text annotations provide absolute precision. The zero baseline is always visible, allowing sign of change to be read immediately.

Alternative encodings rejected. A log-scale line chart was the initial design (see

Section 17). It was rejected because pre-crisis inflation (2–8%) was compressed into an imperceptible band near the axis, violating the expressiveness principle that all data variation must be perceivable. The YoY bar chart solves this by transforming cumulative index values into annual rate-of-change, making both small and large changes visible on the same linear scale.

7.3 Chart 6: Lebanon GDP Relative to Regional Peers

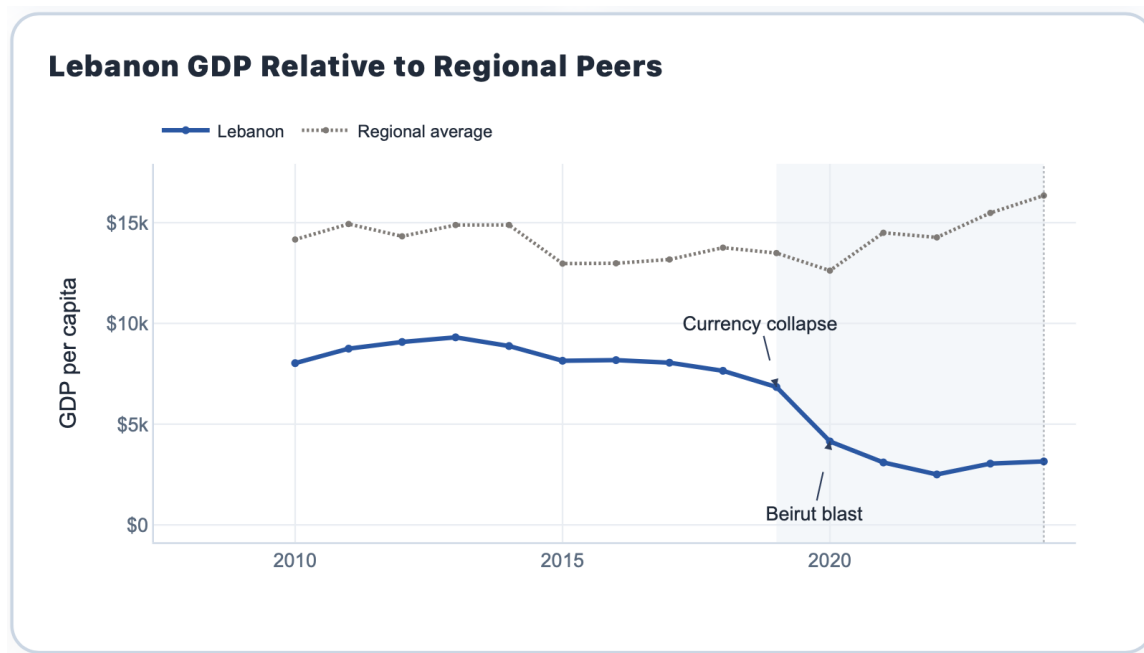


Figure 12: Lebanon GDP per capita relative to regional peers, 2010-2024. Dashed line = regional average; vertical annotation marks the Beirut blast (2020).

What. GDP per capita (USD) for Lebanon (solid line) vs. regional average (dashed line) over Year.

Why. *Discover / Browse / Compare:* track the gap between Lebanon’s GDP and the regional baseline, identifying the collapse inflection point. Target: *trend* and *outlier*.

How - Marks and Channels. *Marks:* two lines. *Channels:* x = Year; y = GDP/cap; line style (solid vs. dashed) = series identity; text annotation marks “Currency collapse” at the inflection year.

Effectiveness. Lebanon’s downward slope post-2019 contrasts sharply with the regional average’s stability, making the divergence the most salient visual feature.

Alternative encodings rejected. A bar chart for each year was rejected because it would make the continuous downward trajectory harder to read than a connected line. A dual-axis chart was rejected to avoid the common misleading comparison problem that

dual axes introduce; the two series share the same unit (USD/capita), so a single shared y-axis is both correct and sufficient.

7.4 Chart 7: Health Financing Composition Over Time (Stacked Area)

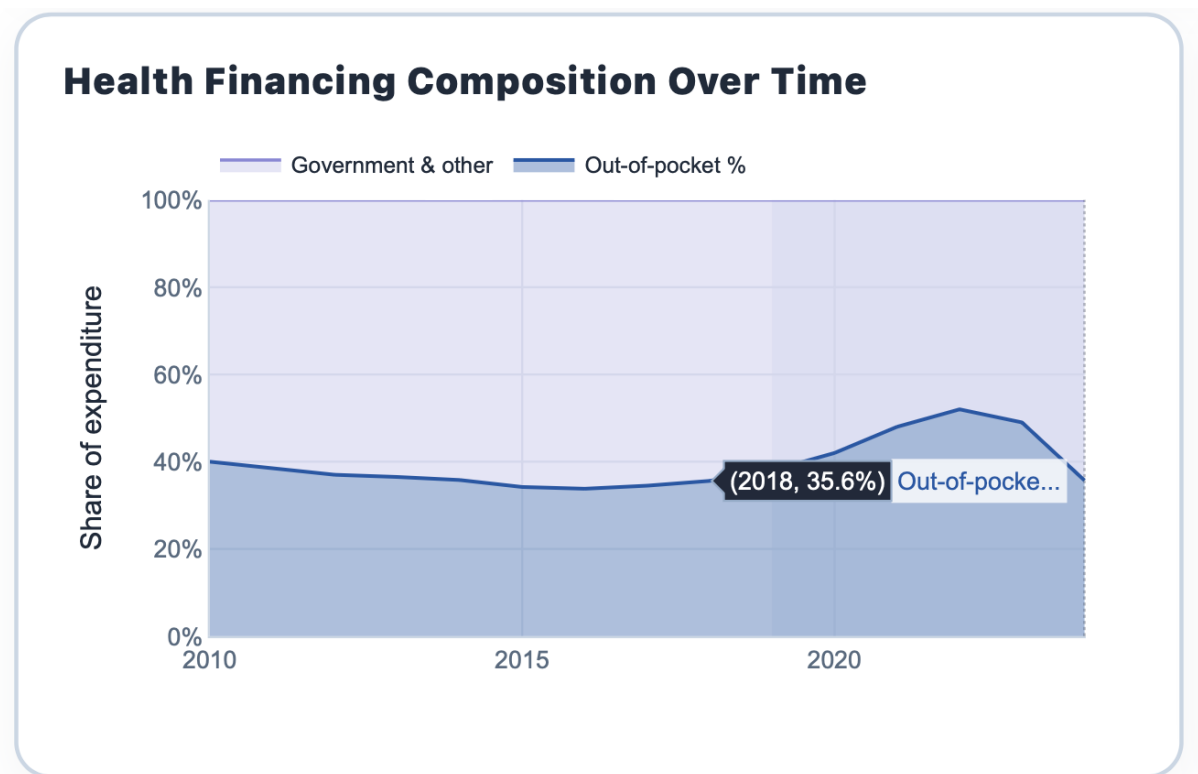


Figure 13: Health Financing Composition Over Time (Lebanon): stacked area showing Out-of-Pocket % (darker) and Government & Other % (lighter), 2010-2024.

What. Two quantitative attributes (OOP % and Government & Other %) summing to 100%, over Year for Lebanon only.

Why. *Discover / Browse / Compare:* track the shifting composition of health financing. Post-2019, OOP share increases as government capacity collapses. Targets: *trend* and *composition*.

How - Marks and Channels. *Marks:* stacked area bands. *Channels:* x = Year; y = cumulative % (0–100); hue = financing source (two-category). The full y-axis span = 100%, providing a fixed proportional frame.

Effectiveness. The expanding OOP band and contracting government band post-2019 are clearly visible, reinforcing the crisis narrative.

Alternative encodings rejected. A line chart for each component was rejected because it would not communicate that the two values are constrained to sum to 100%, which is

the central analytical message. A pie chart per year was rejected because it cannot show temporal change continuously. The stacked area is the correct idiom for part-to-whole composition over time with only two categories.

7.5 Chart 8: Population Size and Migration Pressure

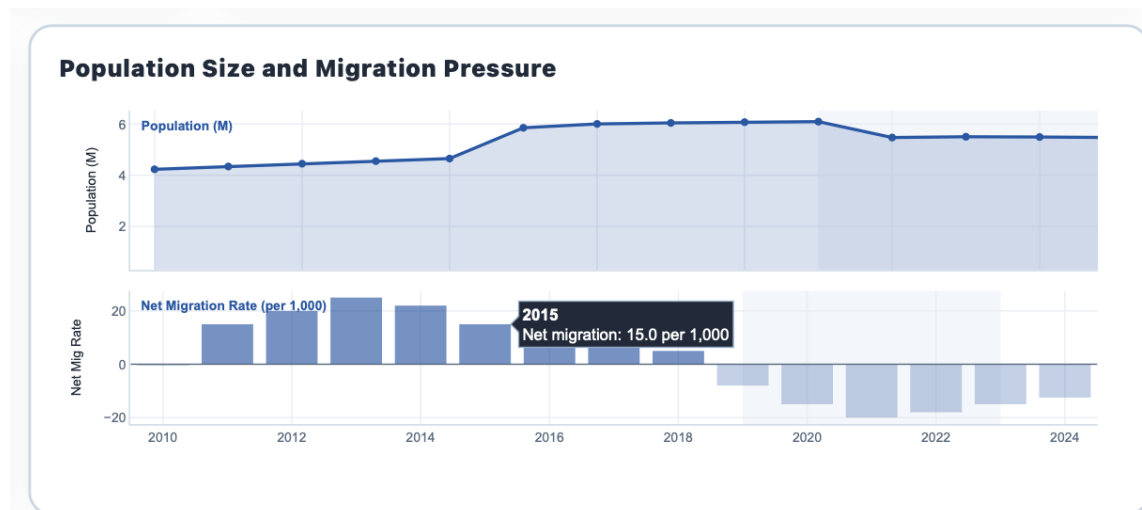


Figure 14: Population Size (top panel, line + area) and Net Migration Rate per 1,000 (bottom panel, bar chart) for Lebanon, 2010-2024. Panels share the same x-axis.

What. Population in millions (sequential) and Net Migration Rate per 1,000 (diverging) for Lebanon over Year.

Why. *Discover / Browse / Compare:* understand whether population stagnation correlates with emigration. Targets: *trends* and *correlation*.

How - Marks and Channels. Two vertically stacked panels sharing the x-axis. *Top panel:* line + shaded area, $y = \text{Population (M)}$. *Bottom panel:* bar chart, $y = \text{Net Migration Rate}$ with explicit zero reference line.

Effectiveness. Stacked panels avoid misleading dual-axis magnitude comparisons while retaining co-variation readability through shared x-axis alignment.

Alternative encodings rejected. A dual-axis line chart was rejected because dual axes are only appropriate when units are the same and can be cognitively misleading when the two series have very different magnitudes. The stacked panel idiom uses the eyes-beat-memory principle: both series are simultaneously visible and the shared x-axis allows direct temporal comparison without requiring the user to remember one chart while looking at the other.

8 Tab 3: Healthcare Expenditure

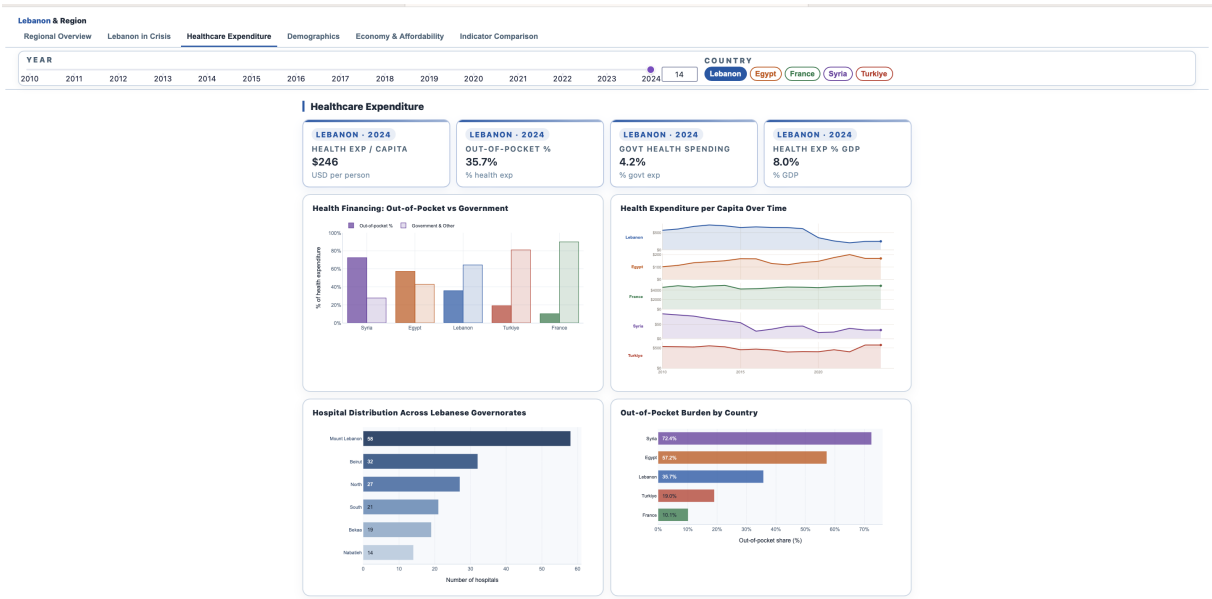


Figure 15: Healthcare Expenditure tab - full view (2024): KPI cards, Health Financing grouped bar chart, and Health Expenditure per Capita Over Time multi-line chart.

8.1 KPI Cards

Card	Value (2024)	Unit	Interpretation
Health Exp / Capita	\$246	USD per person	Per-person spending
Out-of-Pocket %	35.7%	% health exp	Individual burden
Govt Health Spending	4.2%	% govt exp	Government priority
Health Exp % GDP	8.0%	% of GDP	System-level spending

Analysis. The \$246/capita metric reveals Lebanon spends far less per person than France (approximately \$5,000) despite spending 8% of GDP, a paradox explained by Lebanon’s low GDP base. Government spending share fell from approximately 8.5% (2010) to 4.2% (2024), demonstrating fiscal retreat from healthcare.

8.2 Chart 9: Health Financing: Out-of-Pocket vs. Government

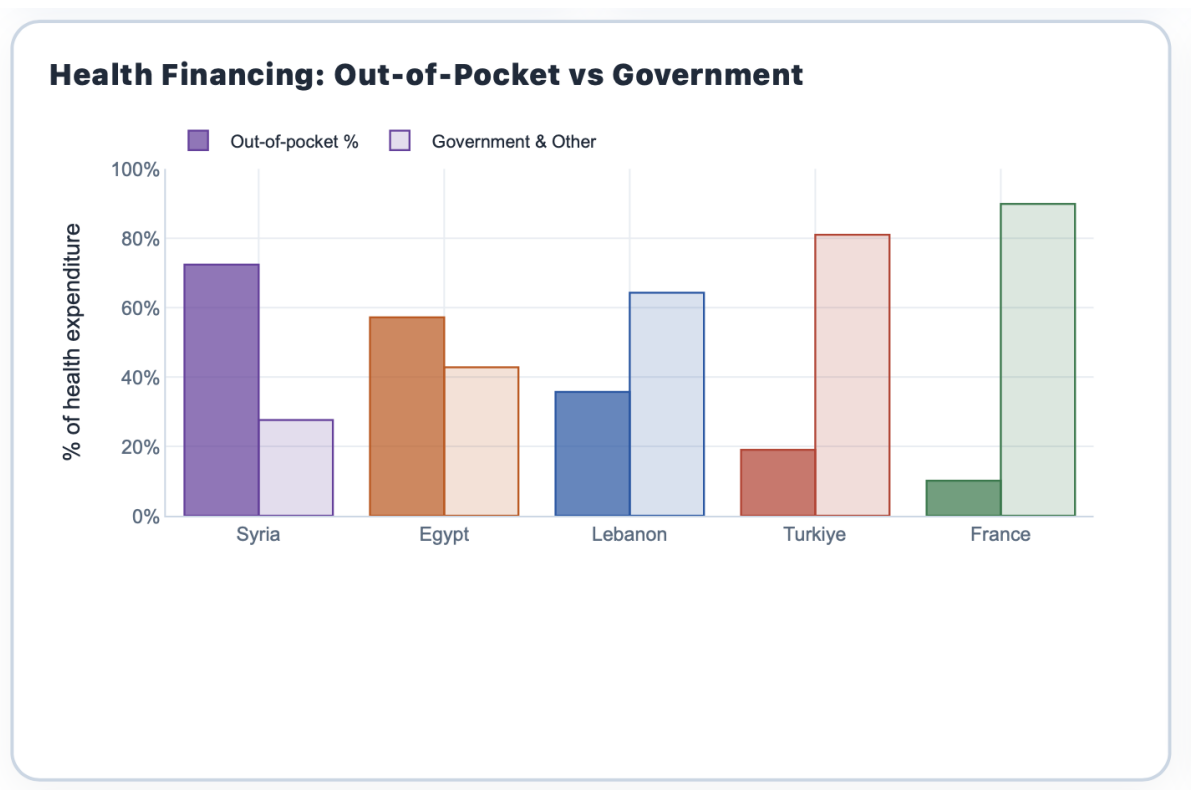


Figure 16: Health Financing: Out-of-Pocket vs. Government, all five countries (selected year). Two bars per country: OOP % (darker) and Government & Other % (lighter/hatched).

What. OOP % and Government & Other % for 5 countries at one selected year.

Why. *Browse / Compare:* rank countries by OOP burden and government commitment simultaneously. Target: *extremes* (Syria 72.4%, France 10.1%) and *distribution*.

How - Marks and Channels. *Marks:* grouped vertical bars (two per country). *Channels:* x = country; bar height = % (aligned length); fill pattern = financing type; hue = country.

Effectiveness. Syria's OOP bar towers over all others. France's low OOP and high government bar illustrate the inverse pattern. **Year-Slider Effect.** Pre-2019 Lebanon shows shorter OOP bar and taller government bar; post-2019 this reverses.

Alternative encodings rejected. A stacked bar chart summing OOP and Government to 100% was considered, but it would hide the fact that the two values already sum to 100% by definition, adding no information over showing them side by side. The grouped layout keeps both bars on the same aligned axis, supporting the highest-ranked channel for magnitude comparison.

8.3 Chart 10: Health Expenditure per Capita Over Time

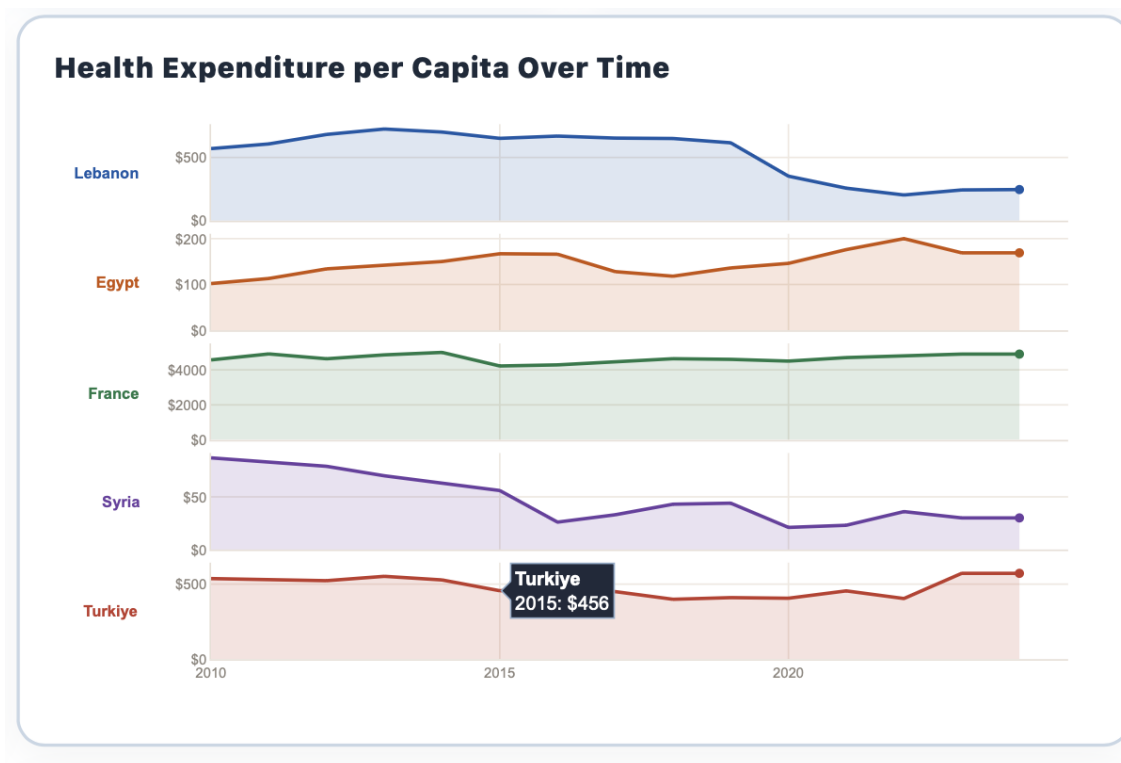


Figure 17: Health Expenditure per Capita Over Time (USD): five countries, 2010-2024. Shaded area bands beneath each line. France leads at approximately \$4,900 per person (2023).

What. Grouped bar chart comparing the Medical Consumer Price Index (CPI) across five countries for the years 2010 and 2024. Temporal and categorical data.

Why. *Compare / Change over time:* observe the inflation of healthcare costs over a 14-year period and compare the magnitude of change between nations. Target: *extremes* (Lebanon’s hyperinflation) and *trends*.

How - Marks and Channels. *Marks:* vertical bars. *Channels:* x = country (categorical); y = CPI value (quantitative, log-like scale suggested by variance); hue/color = year (categorical); data labels = exact index values.

Effectiveness. Side-by-side grouping allows for a direct comparison of “then vs. now.” The inclusion of Lebanon highlights a massive outlier; the use of data labels is critical here because the scale of Lebanon’s increase (from 102 to over 100,000) would otherwise make the metrics for other countries visually disappear if plotted on a standard linear scale.

Alternative encodings rejected. A single shared-axis multi-line chart was the initial design (see Change 5 in Section 17). It was rejected because France’s scale dominated

the y-axis and compressed all other country lines into a flat band near zero. The small multiples approach with independent y-axes was adopted to make each country's internal trend visible.

8.4 Chart 11: Hospital Distribution Across Lebanese Governorates

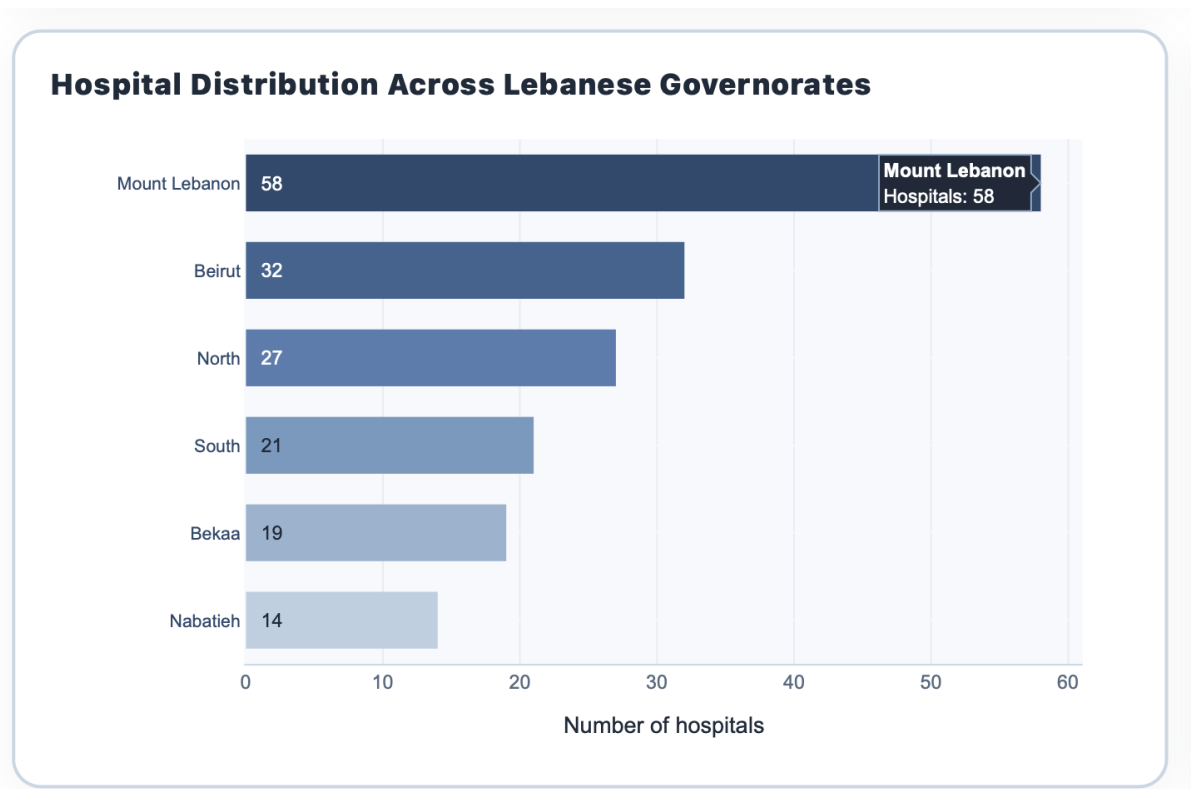


Figure 18: Hospital Distribution Across Lebanese Governorates: horizontal bar chart. Mount Lebanon leads with 58 hospitals; Nabatieh has the fewest (14).

What. Hospital count for 6 governorates. Static data.

Why. *Present / Lookup / Identify:* look up a governorate's hospital count and identify geographic concentration. Target: *extremes* and *distribution*.

How - Marks and Channels. *Marks:* horizontal bars. *Channels:* y = governorate (categorical, spatial region); bar length = count (aligned from zero); luminance = count (redundant); data labels embedded in bars.

Effectiveness. Horizontal orientation allows long governorate names without rotation. Sorted descending order makes ranking immediately apparent.

Alternative encodings rejected. A choropleth map of Lebanon's governorates was considered and explicitly evaluated. It was rejected for the following reasons: (1) there

are only six governorates, which is too few for a choropleth to reveal spatial clustering patterns that the bar chart cannot; (2) the data is raw count, not a rate normalized by area or population, so coloring by count would mislead viewers by conflating hospital density with region size; and (3) the primary task is *rank* and *lookup*, for which aligned bar length is a higher-ranked perceptual channel than color luminance. A map would be appropriate if population-normalized hospital rates were available and geographic pattern discovery were the primary task.

8.5 Chart 12: Out-of-Pocket Burden by Country

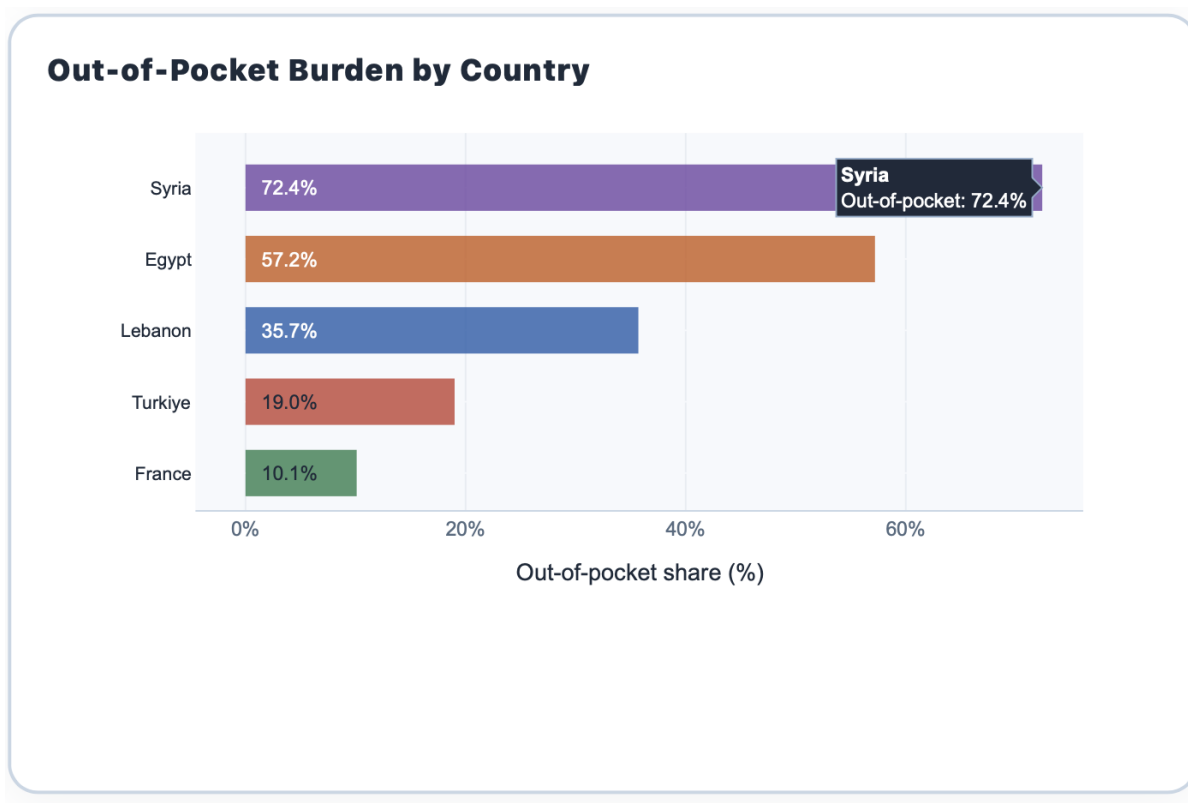


Figure 19: Out-of-Pocket Burden by Country: horizontal bar chart (%). Syria 72.4%; France 10.1%.

What. Out-of-pocket health expenditure for 5 countries. Dynamic data (implied year slider).

Why. *Compare / Rank*: identify countries with the highest individual financial burden. Target: *extremes* and *trends* over time.

How - Marks and Channels. *Marks*: bars. *Channels*: length = percentage; y-axis = country; color = country (identity). Interactive slider enables temporal filtering.

Effectiveness. Descending sort offers instant ranking, though the order and values shift

dynamically based on the selected year. Labels ensure precision.

Alternative encodings rejected. A line chart over time was considered but would require all five countries to be compared across 15 time points simultaneously, producing significant overplotting. The bar chart at the selected year, combined with the interactive slider, allows the user to animate through time step by step, which is more effective for the rank-comparison task than a static multi-line chart.

9 Tab 4: Demographics

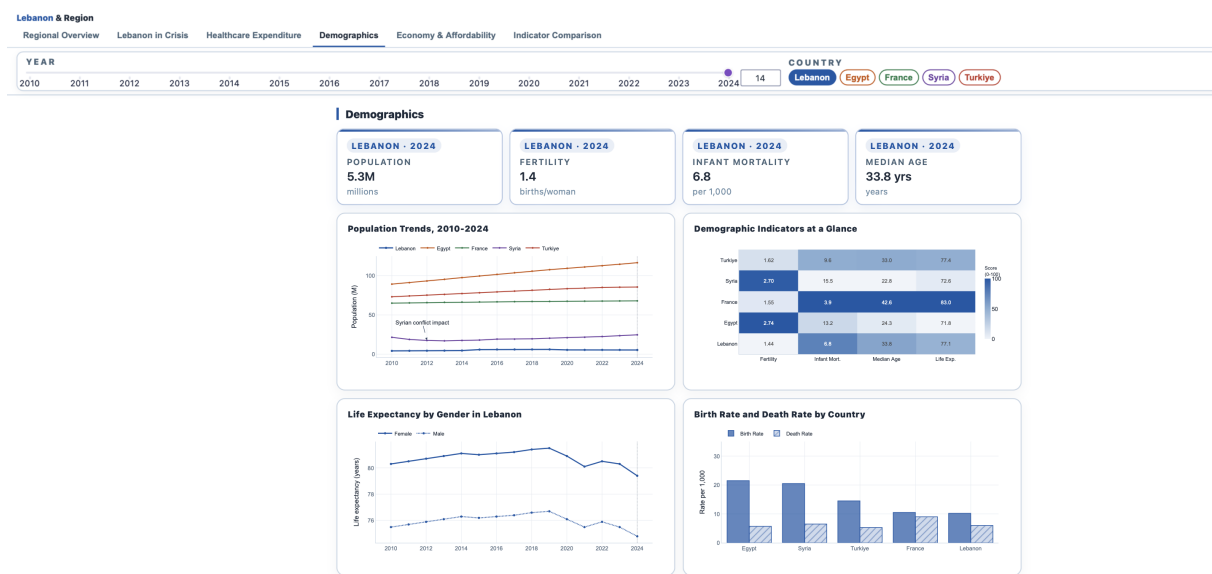


Figure 20: Demographics tab - full view: KPI cards, Population Trends line chart, and Demographic Indicators heatmap table.

9.1 KPI Cards

Card	Value (2024)	Unit	Interpretation
Population	5.3M	millions	Total population
Fertility	1.4	births/woman	Below replacement level
Infant Mortality	6.8	per 1,000	Live births
Median Age	33.8 yrs	years	Aging indicator

Analysis. Lebanon's fertility rate of 1.4 births/woman is below the replacement rate of 2.1, signaling long-term population decline risk compounded by emigration. These KPIs provide demographic context for health system capacity questions.

9.2 Chart 13: Population Trends, 2010–2024

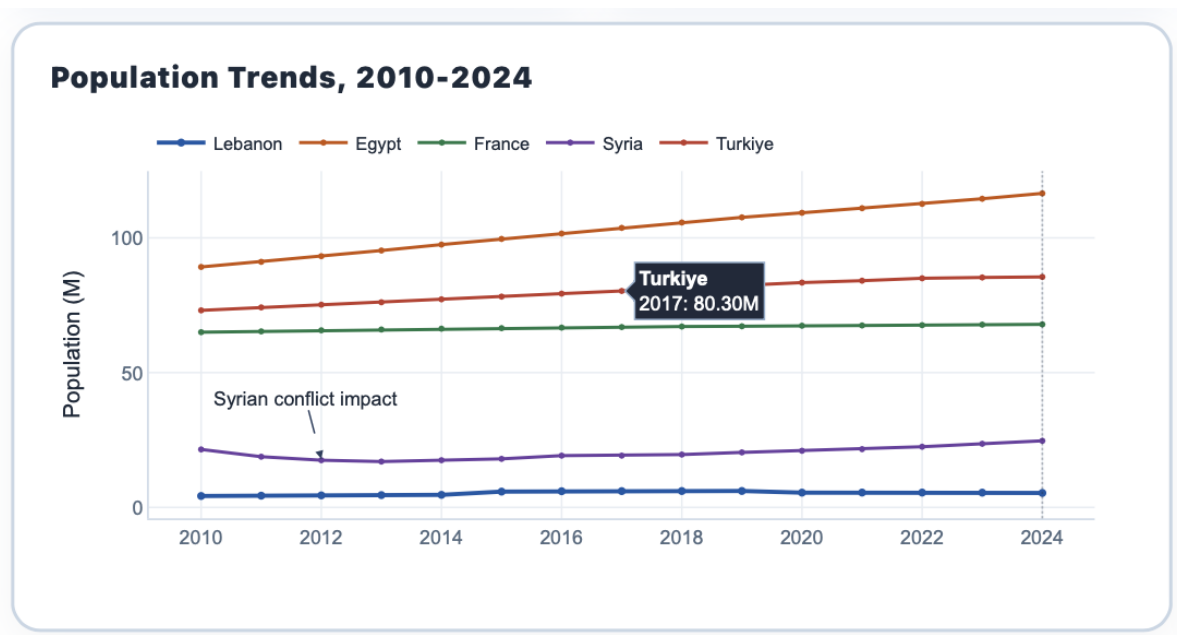


Figure 21: Population Trends, 2010–2024: total population (millions) for all five countries. Egypt and Türkiye trend strongly upward; Syria declines. “Syrian conflict impact” annotation marks the 2011–2012 inflection.

What. Multi-line chart showing population growth for five countries from 2010 to 2024. Temporal data.

Why. *Trends / Comparison / Change over time:* observe population trajectories and compare growth rates between nations. Target: *slopes* (growth speed), *anomalies* (Syrian conflict impact), and *intercepts*.

How - Marks and Channels. *Marks:* lines and points. *Channels:* x = year (quantitative); y = population in millions (quantitative, aligned from zero); hue/color = country (categorical); annotation = contextual event (textual/arrow).

Effectiveness. Line charts are ideal for highlighting continuity and change over time. Using distinct colors and a legend allows for easy differentiation between countries with vastly different population scales, while the specific annotation for Syria provides crucial context for the non-linear trend.

Alternative encodings rejected. A bar chart for each country and year was rejected because it would break temporal continuity, making slope (rate of growth) impossible to read at a glance. An area chart was rejected because filled areas would overlap and obscure the smaller countries (Lebanon, Syria) behind the larger ones (Egypt, Türkiye).

9.3 Chart 14: Demographic Indicators at a Glance (Heatmap)

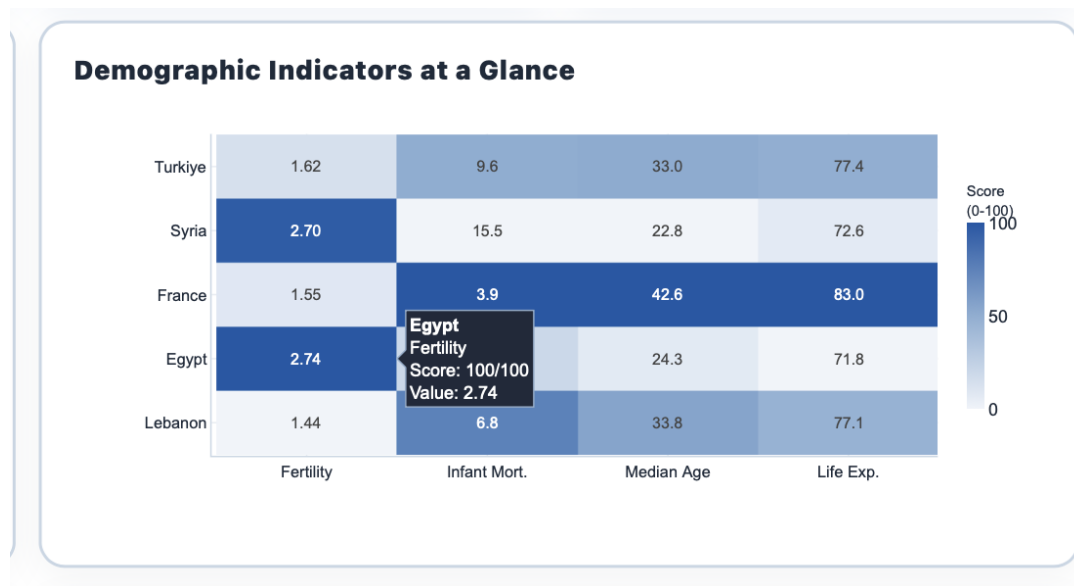


Figure 22: Demographic Indicators at a Glance: matrix heatmap. Rows = countries; columns = Fertility, Infant Mortality, Median Age, Life Expectancy. Color = raw value (sequential blue); text = exact value.

What. Heatmap of four demographic indicators for five countries. Static data.

Why. *Summarize / Compare / Pattern Discovery:* look up specific country metrics and identify correlations between indicators (e.g., high infant mortality vs. low life expectancy). Target: *clusters* and **outliers**.

How - Marks and Channels. *Marks:* area (cells). *Channels:* y = country; x = indicator; luminance/saturation = normalized score (0–100); data labels = raw values. Interactive tooltip displays the underlying normalized score and value.

Effectiveness. The color encoding allows for rapid pre-attentive processing of “best” and “worst” performers across diverse scales (fertility vs. age). Darker blues highlight higher relative scores, though the user must differentiate whether “high” implies a positive or negative societal outcome.

Alternative encodings rejected. Four separate bar charts (one per indicator) were considered but would require the user to move between four views to identify patterns across indicators, which this two-key heatmap idiom reveals in a single glance. The heatmap follows Munzner’s two-key table layout: rows (countries) and columns (indicators) are the two keys, and cell luminance encodes the value.

9.4 Chart 15: Life Expectancy by Gender in Lebanon

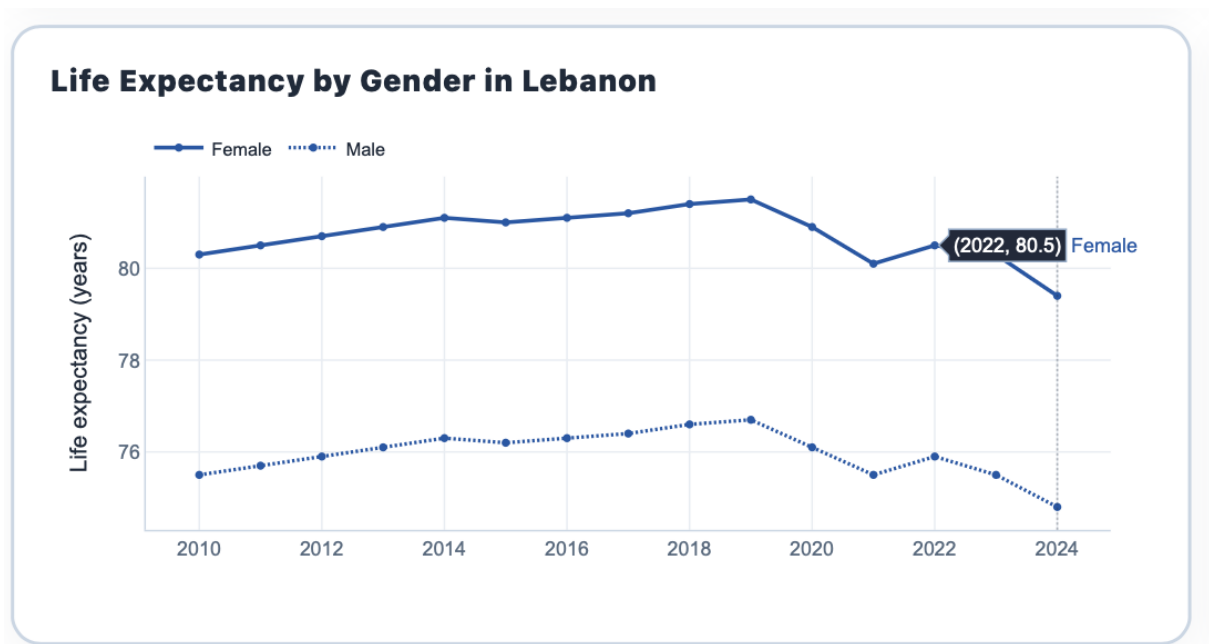


Figure 23: Life Expectancy by Gender in Lebanon, 2010–2024: Female (solid line) consistently higher than Male (dashed line). Dashed vertical line marks 2023 (last confirmed data point).

What. Dual-line chart comparing life expectancy by gender in Lebanon (2010–2024). Temporal data.

Why. *Comparison / Trends:* analyze the gender gap over time and identify recent volatility. Target: *correlations* (parallel dips) and *extremes* (recent declines).

How - Marks and Channels. *Marks:* lines and points. *Channels:* x = year (quantitative); y = years of life (quantitative); hue/color = gender (categorical, single hue); stroke style = gender (solid for female, dotted for male).

Effectiveness. Using a non-zero y-axis start (≈ 74) emphasizes the variance and the specific gap between genders. Redundant encoding (solid vs. dotted) ensures readability even if color is stripped, while the shared trend highlights external shocks affecting both demographics simultaneously.

Alternative encodings rejected. A grouped bar chart per year was rejected because the gender gap is best understood as a continuous temporal relationship rather than a point-in-time comparison. A single bar chart per year showing the gap value was considered, but it would hide the absolute values and prevent the user from judging whether life expectancy itself is rising or falling.

9.5 Chart 16: Birth Rate and Death Rate by Country

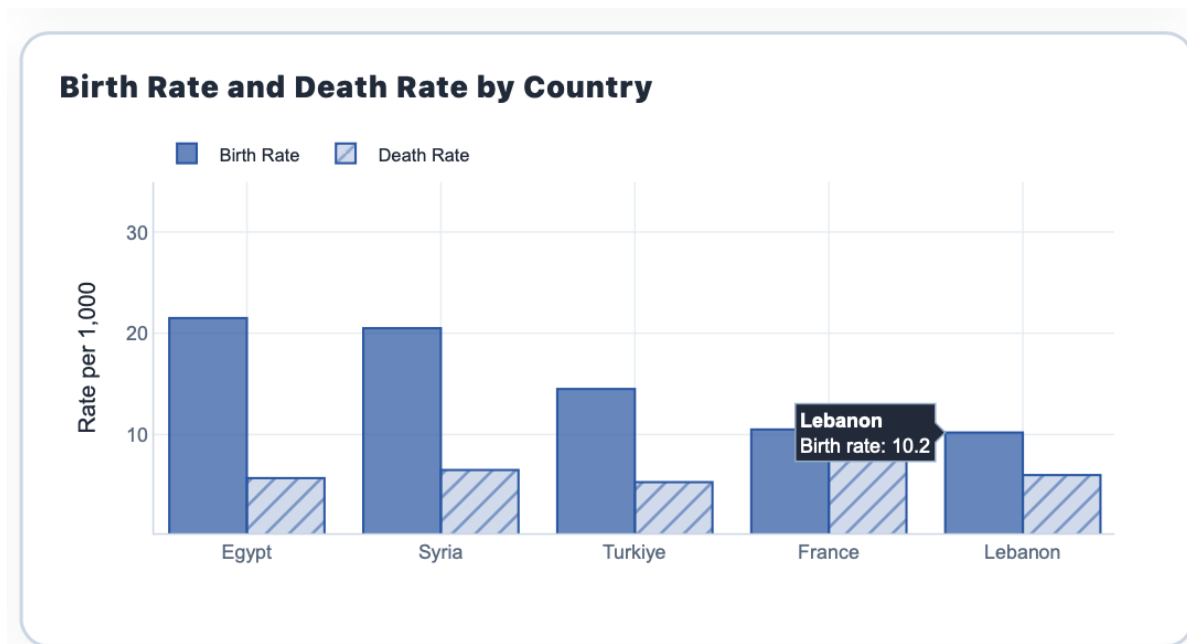


Figure 24: Birth Rate and Death Rate by Country (per 1,000): grouped bars for the selected year. Solid blue = Birth Rate; hatched = Death Rate.

What. Grouped bar chart comparing crude birth and death rates across five countries. Static data.

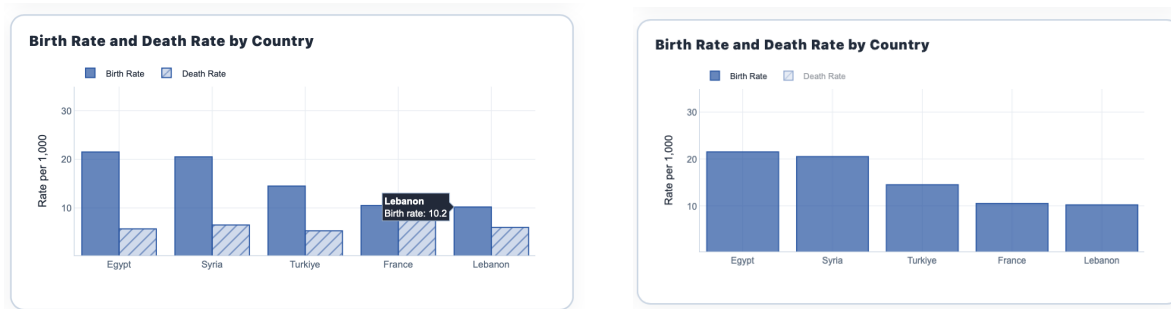
Why. *Compare / Analyze:* evaluate the natural population growth (birth minus death) for each country. Target: *distribution* and *differences* (the gap between rates).

How - Marks and Channels. *Marks:* vertical bars. *Channels:* x = country (categorical); y = rate per 1,000 (quantitative, aligned from zero); hue/pattern = metric type (solid blue for birth, hatched blue for death).

Effectiveness. Grouping bars side-by-side allows for immediate comparison of the two metrics within a single country, making the natural increase rate visually intuitive. The hatched texture on the death rate bars provides a strong redundant cue for accessibility.

Alternative encodings rejected. A stacked bar (birth + death) was rejected because the sum of birth and death rates has no meaningful interpretation. A line chart over time was rejected in favor of the grouped bar because the primary task is cross-country comparison at a given year, not temporal trend, and the year slider provides the temporal dimension interactively.

9.5.1 Legend Toggle Interaction: Before / After



(a) Both Birth Rate and Death Rate visible for all countries.

(b) Death Rate hidden through legend interaction, allowing clearer comparison of Birth Rates only.

Figure 25: Birth Rate and Death Rate chart - legend toggle interaction effect.

Interactive Filtering. The chart supports legend-based filtering, allowing users to temporarily hide one variable and focus on the remaining data. In the second view, the Death Rate bars are hidden, making Birth Rate comparisons across countries easier to interpret. This interaction reduces visual clutter and supports focused analytical exploration.

10 Tab 5: Economy & Affordability

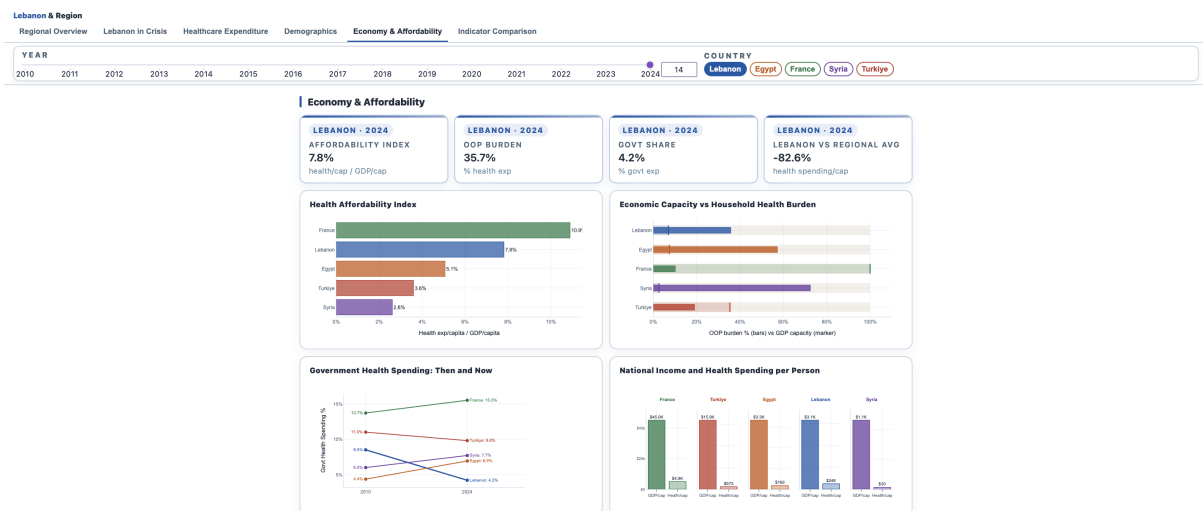


Figure 26: Economy & Affordability tab - full view: KPI cards, Health Affordability Index bar chart, and Economic Capacity vs Household Health Burden chart.

10.1 KPI Cards

Card	Value (2024)	Unit	Interpretation
Affordability Index	7.8%	health/cap ÷ GDP/cap	Derived ratio
OOP Burden	35.7%	% health exp	Individual burden
Govt Share	4.2%	% govt exp	Government priority
Lebanon vs Regional Avg	−82.6%	health spending/cap	Gap vs. regional mean

Analysis. The Affordability Index ($7.8\% = \text{Health Exp/cap} \div \text{GDP/cap}$) normalizes health spending by income. France scores 10, Lebanon 7.8%, Egypt 5.1%. Lebanon allocates a proportionally higher income share to health than Egypt despite larger absolute OOP burden. The -82.6% card signals Lebanon's per-capita health spending is now 82.6% below the regional mean.

10.2 Chart 17: Health Affordability Index

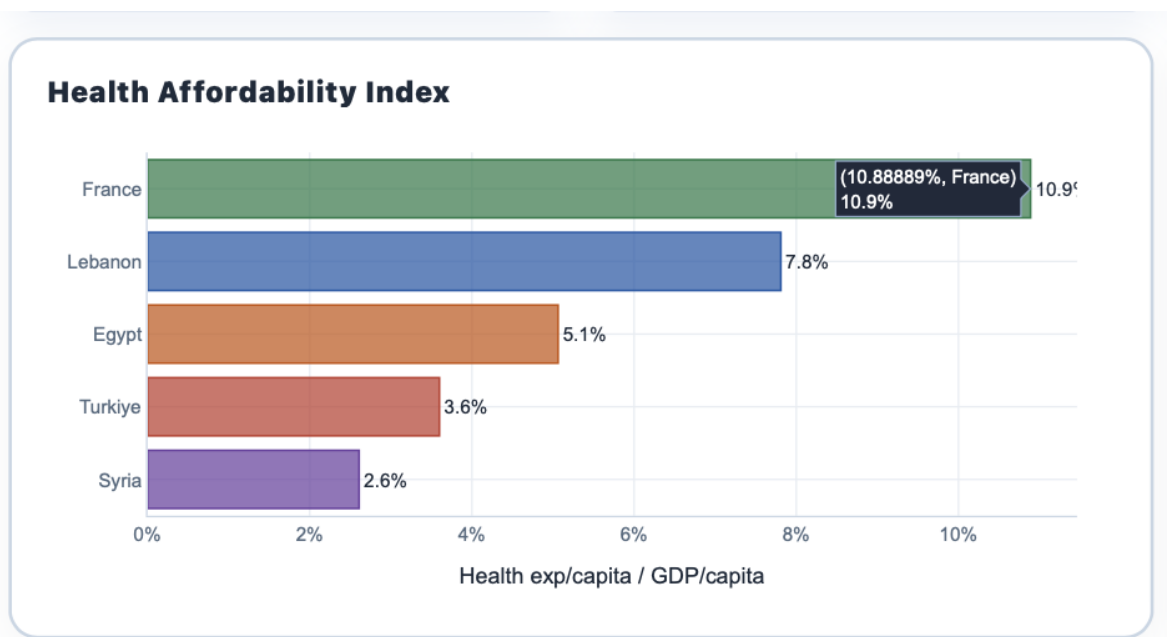


Figure 27: Health Affordability Index: Health Exp per capita / GDP per capita for five countries. France leads at 10%; Syria trails at 2.6%. Sorted descending.

What. Health Affordability Index (ratio of health expenditure per capita to GDP per capita) for five countries. Static data.

Why. *Present / Rank:* evaluate the relative economic weight of healthcare costs across different economies. Target: *extremes* (France vs. Syria) and *comparison* of healthcare investment intensity.

How - Marks and Channels. *Marks:* horizontal bars. *Channels:* y = country (categorical); bar length = index percentage (quantitative, aligned from zero); hue/color = country (categorical); data labels = exact percentage.

Effectiveness. The horizontal orientation accommodates country names clearly. The descending sort order highlights the disparity in how much of a country's average economic output per person is dedicated to health, with France appearing as the most intensive investor.

Alternative encodings rejected. A scatter plot of the two component values (Health Exp/cap and GDP/cap) was considered but would require the user to mentally compute the ratio rather than reading it directly. Since the derived ratio is itself the meaningful analytical attribute, encoding it directly as bar length is more expressive and effective.

10.3 Chart 18: Economic Capacity vs. Household Health Burden

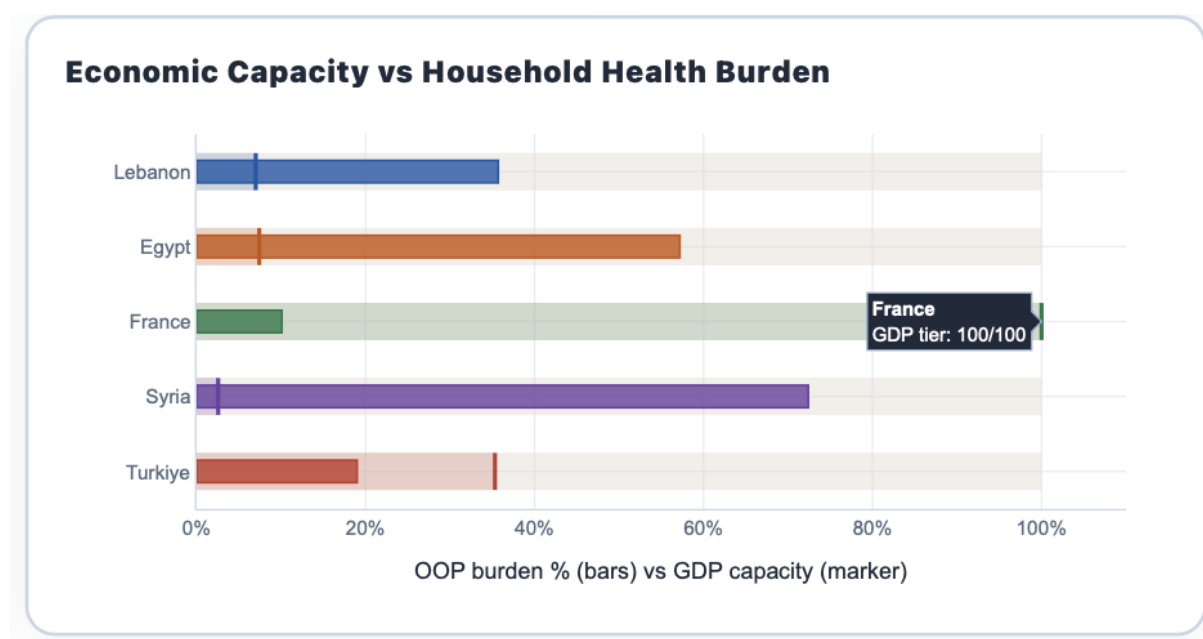


Figure 28: Economic Capacity vs Household Health Burden: horizontal bars = OOP burden (%); point markers = GDP capacity (relative position). x-axis spans 0–100%.

What. Bullet chart comparing household out-of-pocket (OOP) burden against economic capacity (GDP tier) for five countries. Static data.

Why. *Compare / Evaluate:* assess the relationship between a country's economic strength and the financial burden placed on its citizens. Target: *disparities* and *performance gaps* (e.g., high burden despite low capacity).

How - Marks and Channels. *Marks:* horizontal bars (burden) and vertical tick marks

(capacity). *Channels*: y = country; bar length = OOP burden %; tick position = GDP tier (0–100 scale); hue = country (categorical); background shaded area = range/capacity context.

Effectiveness. The bullet chart idiom is highly efficient for comparing a primary metric against a benchmark. By overlaying the GDP “target” tick onto the OOP “actual” bar, the visualization highlights systemic imbalances such as Syria’s high burden exceeding its low economic capacity mark.

Alternative encodings rejected. The original scatter plot design (see Change 2 in Section 17) required simultaneous estimation of two axis positions to identify country performance, which is cognitively demanding for a comparison task. The bullet chart replaces the second axis with a single tick mark, freeing the primary axis for aligned length comparison, a higher-ranked perceptual channel.

10.4 Chart 19: Government Health Spending: Then and Now (Slope Chart)

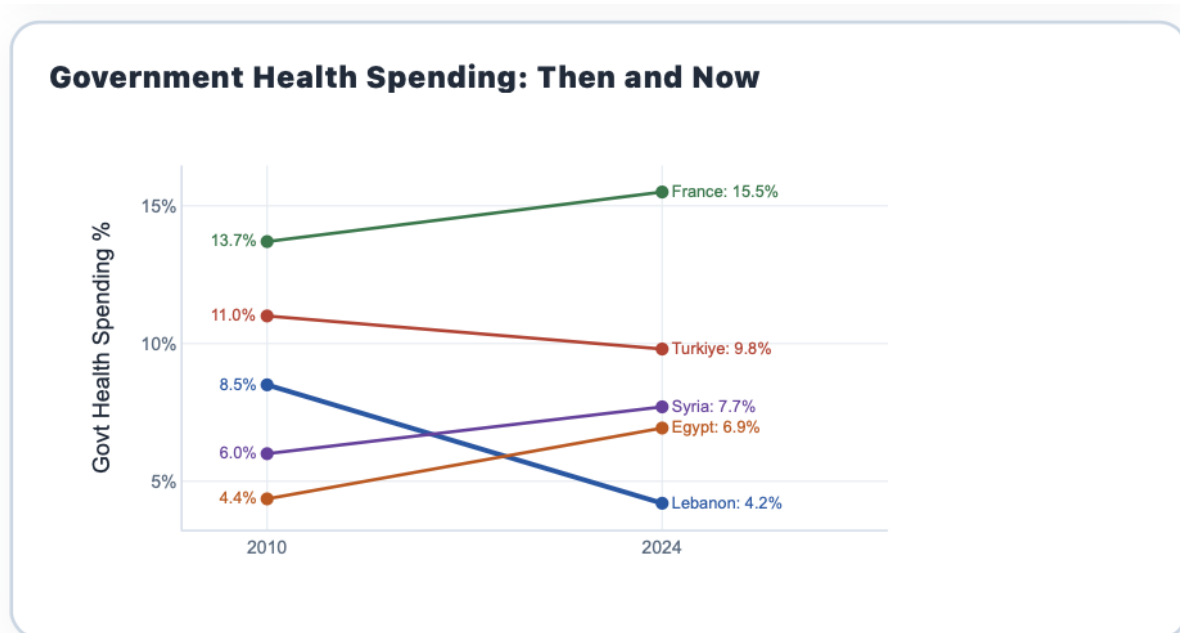


Figure 29: Government Health Spending: Then and Now. Slope chart comparing 2010 (left) vs. 2024 (right) for all countries. France: 15.5%, Türkiye: 9.8%, Syria: 7.7%, Egypt: 6.9%, Lebanon: 4.2%.

What. Govt Health Spending (% of govt exp) for 5 countries at two time points: 2010 and 2024.

Why. *Discover / Browse / Compare*: show direction and magnitude of change in government health prioritization. Target: *trends* as two-point trajectories.

How - Marks and Channels. *Marks:* points and slope lines. *Channels:* x = time point (binary categorical); y = % of govt exp (aligned position); hue = country; slope steepness and direction = change magnitude and direction (derived channel).

Effectiveness. Lebanon's steep downward slope from $\approx 8.5\%$ (2010) to 4.2% (2024) is the most dramatic. France's nearly flat line at 15% confirms stable prioritization. Country labels at right endpoint eliminate legend look-up.

Alternative encodings rejected. A full multi-line time series (see Change 3 in Section 17) was rejected because overlapping lines across 15 years make the direction of net change difficult to read. The slope chart reduces this to the minimum: two endpoints and the direction of movement between them, which is precisely the analytical question being asked.

10.5 Chart 20: National Income and Health Spending per Person

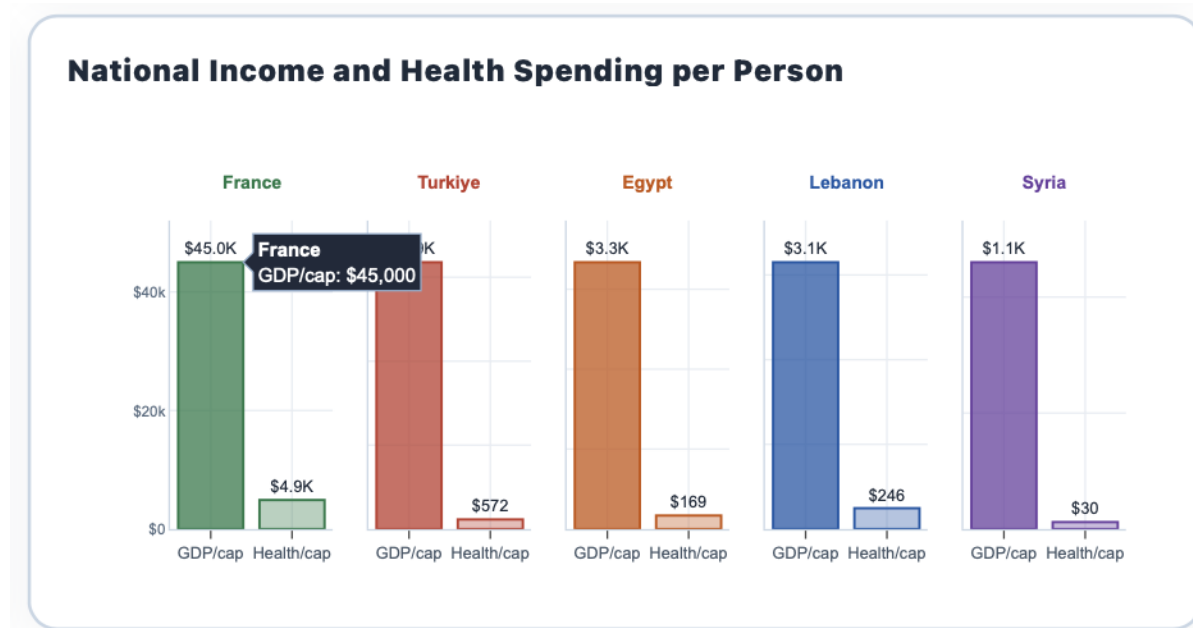


Figure 30: National Income and Health Spending per Person: grouped vertical bars comparing GDP per capita and Health Expenditure per capita across all countries for the selected year.

What. GDP per capita and Health Expenditure per capita (USD) for five countries at the selected year.

Why. *Present / Lookup / Compare:* expose absolute differences in economic capacity and healthcare spending resources across countries. The chart highlights disparities in both wealth and healthcare investment.

How - Marks and Channels. *Marks:* grouped vertical bars. *Channels:* x = country

(categorical); y = USD value (aligned vertical position); fill shade = metric type (GDP per capita vs. Health Expenditure per capita); text labels provide precise numerical values.

Effectiveness. France's GDP bar dominates the chart, visually separating it from the regional countries. Lebanon's health expenditure per capita remains substantially lower than France despite maintaining a relatively high Health Expenditure % GDP. Syria's extremely small Health Expenditure per capita bar reveals severe healthcare resource limitations.

Alternative encodings rejected. A ratio or index was considered as an alternative, but the absolute USD values are necessary to convey the real-world magnitude of the spending gap. A scatter plot was rejected because the task is cross-country comparison of two discrete values per country, not correlation discovery.

11 Tab 6: Indicator Comparison

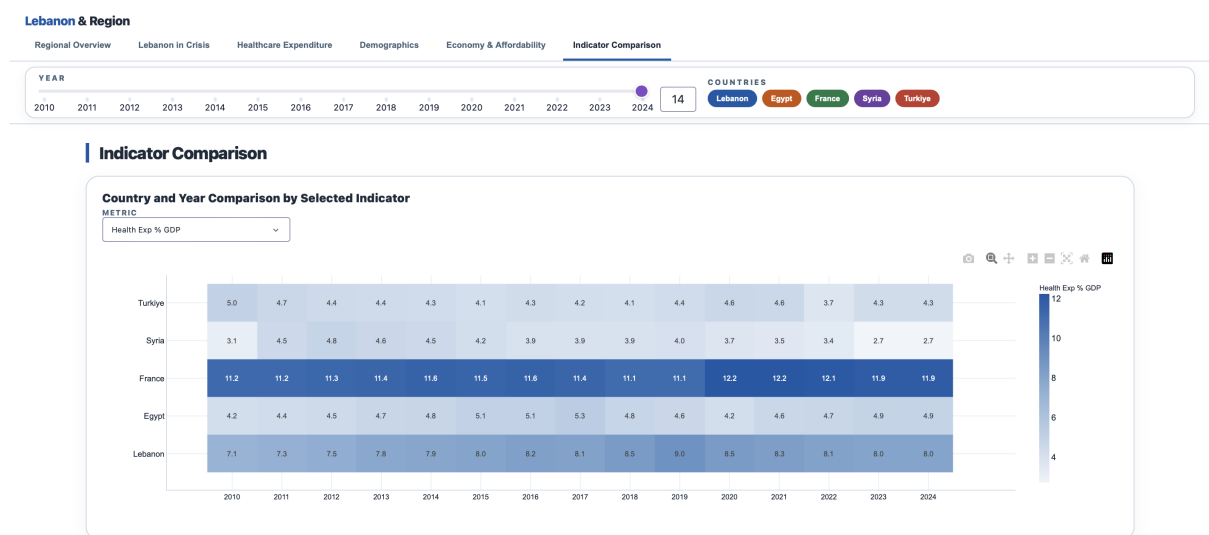


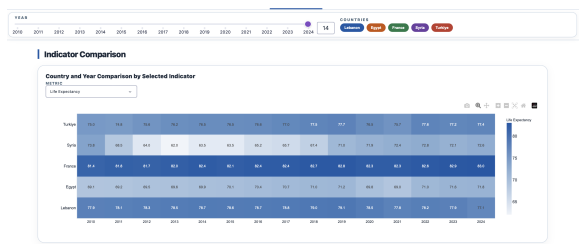
Figure 31: Indicator Comparison tab - full view: interactive heatmap comparing countries and years for a selected indicator using the metric dropdown and country slicer controls.

11.1 Country Slicer Interaction

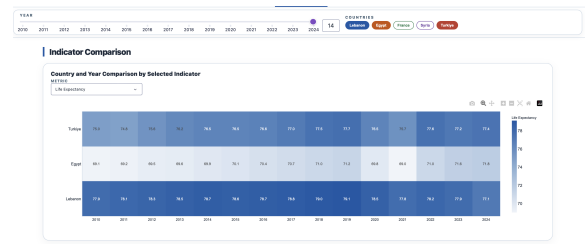
What. Country filter buttons allow users to dynamically remove or restore countries from the heatmap while preserving the selected metric.

Why. This interaction supports filter operation by reducing visual clutter and enabling focused comparison among selected countries.

How. Clicking a country button updates the Dash callback, filters the dataset, and redraws the heatmap with only active countries.



(a) Life Expectancy heatmap with all five countries visible.



(b) France and Syria removed using the country slicer, allowing clearer comparison among the remaining countries.

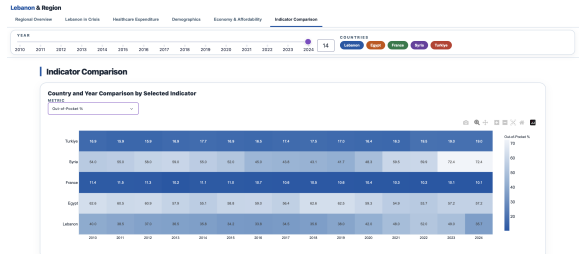
Figure 32: Country slicer interaction effect on the heatmap visualization.

11.2 Metric Dropdown Interaction

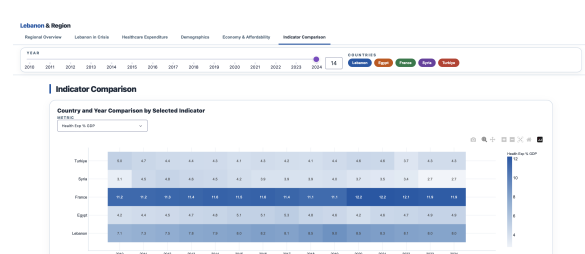
What. The “METRIC” dropdown allows users to switch between multiple indicators while preserving the same heatmap structure.

Why. This interaction supports encode operation by changing the attribute mapped to color luminance without changing the visualization idiom.

How. Selecting a new metric triggers a Dash callback that updates the dataset, rescales the color mapping, and redraws the heatmap.



(a) Metric set to Out-of-Pocket %. Syria contains the darkest cells due to the highest burden values.



(b) Metric changed to Health Expenditure % GDP. France becomes visually dominant due to consistently higher values.

Figure 33: Metric dropdown interaction effect on the heatmap visualization.

12 Marks and Channels Summary

Following Munzner’s channel ranking: *aligned position* > *unaligned position* > *length* > *angle* > *area* > *luminance*:

Chart	Mark	Primary Channels	Tab
Health Exp % GDP (line)	Line+Point	x=Year, hue=country	y=%GDP, Overview

(continued)

Chart	Mark	Primary Channels	Tab
Income & Life Exp	Point	x=GDP/cap, y=Life Exp, hue=country	Overview
Radar (profiles)	Area+Line	radius=score, angle=indicator, hue=country	Overview
CPI Bar (YoY)	Bar	x=Year, height=%change, text=label	Crisis
Lebanon GDP (line)	Line	x=Year, y=GDP/cap, style=series	Crisis
Health Financing (area)	Area	x=Year, y=%, hue=source	Crisis
Pop+Migration (dual)	Line+Bar	x=Year, y1=pop, y2=mig rate	Crisis
OOP vs Govt (grouped bar)	Bar	x=country, height=%, pattern=type	Healthcare
Health Exp/cap (multi-line)	Line+Area	x=Year, y=USD, hue=country	Healthcare
Hospital Bar	Bar	y=governorate, length=count, lum=count	Healthcare
OOP by Country	Bar	y=country, length=OOP%, hue=country	Healthcare
Pop Trends (line)	Line+Point	x=Year, y=pop(M), hue=country	Demographics
Demo Heatmap	Area(cell)	x=indicator, y=country, lum=value	Demographics
Life Exp Gender	Line	x=Year, y=years, style=gender	Demographics
Birth/Death Bar	Bar	x=country, height=rate, pattern=type	Demographics
Affordability Bar	Bar	y=country, length=ratio, hue=country	Economy
OOP vs GDP (composite)	Bar+Point	length=OOP%, x-pos=GDPcap	Economy
Govt Spending (slope)	Point+Line	x=endpoint, y=%, slope=change	Economy

(continued)

Chart	Mark	Primary Channels	Tab
National (grouped bar)	Income Bar	x=country, height=USD, shade=metric	Economy
Indicator Heatmap	Area(cell)	x=Year, y=country, lum=value	Indicator

12.1 Redundant Encoding

In all multi-country line charts, country identity is encoded by both **color hue** and **marker shape**. In bar charts, values are encoded by both **bar length** and **embedded text labels**. In heatmaps, values are encoded by both **color luminance** and **text label**. These redundancies improve accessibility and support both gestalt and precise query tasks.

13 Color Mapping

13.1 Country Identity Colors

Country	Color	Role
Lebanon	Blue (#1558A8)	Focal country
Egypt	Orange (#FF7F0E)	Regional peer
France	Teal/Green (#2CA02C)	European benchmark
Syria	Purple (#9467BD)	Regional peer
Türkiye	Red (#D62728)	Regional peer

Following Munzner's color channel taxonomy, hue is used here as a categorical identity channel, which is its correct application: hue has no perceptual ordering and therefore should not be used to encode magnitude. Hues are drawn from distinct perceptual segments of the color wheel, maximizing discriminability. The mapping is consistent across all 20+ charts so users build a stable mental model. Blue/orange and blue/red pairs are colorblind-safe; the green/red pair (France/Türkiye) relies on redundant shape encoding for accessibility.

13.2 Sequential Colormaps

Heatmap cells: light blue (low) to deep blue (high). Single-hue sequential colormaps are perceptually monotone (luminance varies) and colorblind-safe. **Hospital bar chart:** light-to-dark blue encoding hospital count. Both avoid implying false categorical group-

ings. A diverging colormap was not required for these views because all encoded values in the heatmaps are sequential (no meaningful zero crossing). The two diverging attributes in the dataset (Net Migration Rate and Pop Growth Rate) are represented using bar charts with an explicit zero reference line, which is the appropriate idiom for diverging quantitative data.

14 Interaction Design

The dashboard implements Munzner’s four interaction idiom categories:

Navigate. Tab navigation between the six thematic views. Plotly’s native toolbar (zoom, pan, reset) enables fine-grained chart exploration.

Filter. Year slider filters all point-in-time charts (scatter plots, bar charts, KPI cards, radar). Country toggle buttons filter multi-country charts. Plotly legend-item clicks filter individual series within charts.

Encode. Metric dropdown on Tab 6 swaps the encoded attribute in the heatmap matrix, changing what quantity maps to color luminance without altering the idiom.

Detail on Demand. Tooltips reveal precise numerical values on hover, compensating for imprecision of angle (radar), area (bubble), and luminance (heatmap) channels.

14.1 Cross-Tab Linking

The year slider and country toggle buttons apply globally across all tabs, a form of **brushing-and-linking** where the filter state is the “brush”. Within-chart selections do not currently propagate to other charts; adding Dash callback chains for cross-chart country-selection filtering would further enhance exploratory analysis.

15 Rules of Thumb Application

This section applies the visualization rules of thumb to the specific design decisions made in the dashboard.

No unjustified 3D. All 20 charts use 2D planar encodings. No 3D bars, 3D pie charts, or perspective projections are used anywhere in the dashboard. The power of the plane (x and y aligned spatial position) is preserved throughout, ensuring that the highest-ranked perceptual channels are always available for magnitude encoding. Adding a third depth dimension would introduce perspective distortion and occlusion without adding any analytical value for this dataset.

Eyes beat memory. The juxtaposed tab structure places related charts side by side within each tab so that users move their eyes between charts rather than holding one chart in memory while navigating to another. The use of small multiples for the Health Expenditure per Capita chart (Change 5) applies this principle directly: showing all five countries simultaneously on faceted panels is more effective than animating through each country sequentially. The global year slider and country toggle buttons update all charts simultaneously so that comparisons do not require memorizing a prior state.

Overview first, zoom and filter, details on demand. The dashboard follows this principle at multiple levels. The Regional Overview tab provides the macro summary. The year slider and country buttons filter the data space. Individual chart tooltips provide details on demand by revealing precise numerical values that are too fine-grained to encode in the primary visual channel. This hierarchy maps directly to Shneiderman's mantra, which is the practical application of Munzner's rule.

Responsiveness is required. All Dash callbacks are configured to update only the figures that depend on the changed control, rather than redrawing the entire dashboard. KPI cards re-render in under one second on slider drag. Tooltips appear on hover without any perceptible delay, keeping interaction within the subsecond response threshold required for perceptual processing.

Function first, form next. Every design iteration in Section 17 was driven by a functional failure: the log-scale CPI chart failed to make pre-crisis variation perceivable; the scatter plot failed to support direct comparison of OOP burden; the shared-scale multi-line chart failed to show internal trends for lower-income countries. Aesthetic choices such as color palette and card styling were applied after the functional encoding decisions were finalized.

Resolution over immersion. The dashboard uses standard 2D chart idioms rendered at full screen resolution rather than any immersive or 3D presentation. This ensures that axis labels, tick marks, and data annotations are legible and that the user can read precise values without cognitive effort from occlusion or perspective distortion.

16 Tables, Spatial Data, and Networks

16.1 Arranging Tables

Following keys-and-values framework, the datasets in this dashboard contain attributes that serve as keys (independent indices used to look up items) and attributes that serve as values (dependent measurements). The flat Country x Year table has two keys and

multiple values, which classifies it as a two-key multidimensional table.

The primary table idioms used are: (1) the two-key matrix heatmap in Tabs 4 and 6, which places one key (country) on rows and one key (year or indicator) on columns, with luminance encoding the value; (2) the one-key horizontal bar charts, which place one key (country or governorate) on the y-axis and encode a single value as bar length; and (3) the zero-key scatter plots, which express two quantitative values against each other with no categorical key on either axis.

These choices match the data structure: two-key data goes into heatmaps, one-key data goes into bar charts, and two quantitative values go into scatter plots. This alignment between data key structure and idiom type satisfies the expressiveness principle.

16.2 Spatial Data and Maps

The dataset contains one attribute with an explicit geographic dimension: hospital counts by Lebanese governorate. A choropleth map was considered and evaluated as an alternative to the horizontal bar chart (Chart 11). It was ultimately rejected for the following reasons.

First, the data is raw count (number of hospitals), not a rate normalized by area or population. Coloring governorate polygons by raw count would conflate hospital density with region area, producing a misleading impression that larger-area governorates (Bekaa, South) are better served than they are relative to their populations. A choropleth is most defensible when the encoded variable is already a rate or density, such as hospitals per 100,000 residents.

Second, there are only six governorates. With so few spatial units, a choropleth provides no perceptual advantage over a bar chart: the user reads six color values rather than six bar lengths, and color luminance is a lower-ranked channel than aligned length for magnitude comparison.

Third, the primary task is rank and lookup (which governorate has the most hospitals), for which descending sorted bars are optimal. A map reorders governorates by geography rather than by value, making the ranking task harder.

A spatial map would be appropriate in a future iteration that incorporates population-normalized hospital rates, sub-governorate data, or patient travel distance metrics, where geographic pattern discovery becomes a primary task.

16.3 Networks and Trees

Network and tree visualizations are not applicable to this dataset and have been intentionally excluded. The five countries in the dashboard (Lebanon, Egypt, France, Syria, Türkiye) are independent observational units. There is no relational structure between them: no country is a parent, neighbor, or connected node of another in any analytically meaningful sense for this study. Imposing a node-link diagram would require fabricating links between countries, which would encode false topology and mislead users about the nature of the data.

Network visualization would be appropriate for a future extension of this project that incorporates, for example, healthcare worker migration flows between countries (a directed weighted network), referral relationships between hospitals within Lebanon (a node-link tree or graph), or policy influence networks among regional health ministries. In the current dataset, the relational structure between entities is entirely captured by the shared Country and Year key dimensions, which are better represented by the juxtaposed, coordinated multi-view approach already in use.

17 Design Validation and Changes

17.1 Validation Across the Four Nested Levels

Domain-level validation addresses the threat of targeting the wrong users or mischaracterizing their questions. The target users (health economists, policymakers, journalists) were identified through domain knowledge of Lebanon’s public health crisis, and the four high-level analytical questions (Section 1.3) were formulated to reflect the questions that these users would realistically bring to the data. The dashboard subtitle annotations and KPI card labels were designed to make the interface legible to non-specialist users, mitigating the audience mismatch threat.

Abstraction-level validation addresses the threat of mapping domain data and tasks to the wrong abstract types. Data types were verified against taxonomy (categorical, ordinal, quantitative sequential, quantitative diverging) as documented in Section 2.2. Task types were mapped to the {action, target} pair vocabulary (Section 3). The decision to encode the Medical CPI as a year-over-year percentage change rather than the raw cumulative index is an example of abstraction-level intervention: the raw index value does not answer the user’s question (“how fast are prices rising?”); the derived rate does.

Idiom-level validation addresses the threat of choosing an incorrect or ineffective visual encoding or interaction design. This is the level at which the five iterative redesigns

described below were conducted. Each change was motivated by an expressiveness or effectiveness failure identified through visual inspection and reasoning from perceptual principles. Expressiveness was checked by verifying that every data variation in the encoded attribute is perceivable, and that no false patterns are implied. Effectiveness was checked by confirming that the highest-ranked appropriate channels were used for the most important attributes.

Algorithm-level validation addresses the threat of incorrect or slow computation. All data was pre-aggregated at the country-year level before dashboard load, reducing per-interaction computation to simple filtering operations. Normalization scores for the radar chart were computed once at load time and stored in memory. Log-transformed CPI values were replaced by derived YoY change values to avoid log-scale misinterpretation at the display level.

17.2 Change 1: Medical CPI - YoY Bar Chart replacing Log-Scale Line

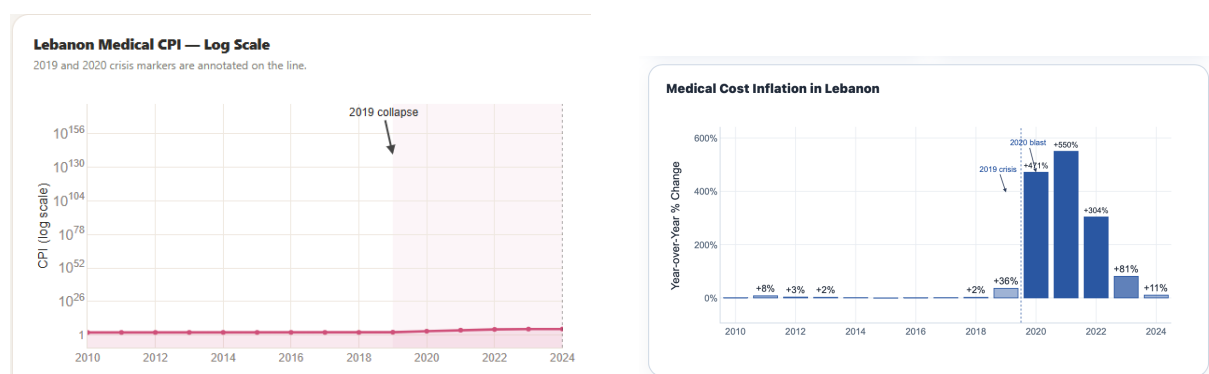
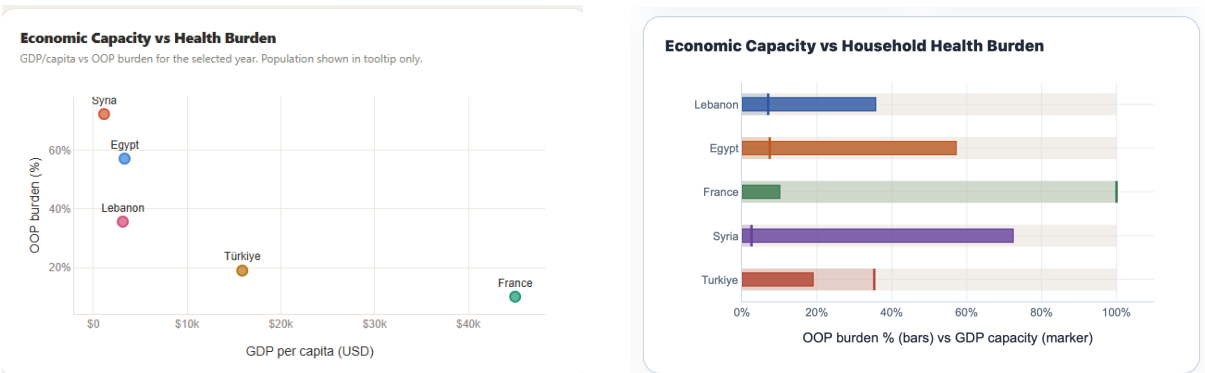


Figure 34: CPI chart redesign: replacing a logarithmic-scale line chart with a year-over-year bar chart to improve interpretability and visibility of inflation changes.

Justification. The YoY % change directly answers “how fast are medical prices rising?” without requiring logarithmic scale interpretation. Bar length from a zero baseline satisfies expressiveness for a ratio-change attribute. Expressiveness principle requires all data variation to be perceivable; the new chart makes gradual pre-crisis inflation visible alongside the post-2019 hyperinflation.

17.3 Change 2: Redesigning Economic Capacity vs Health Burden

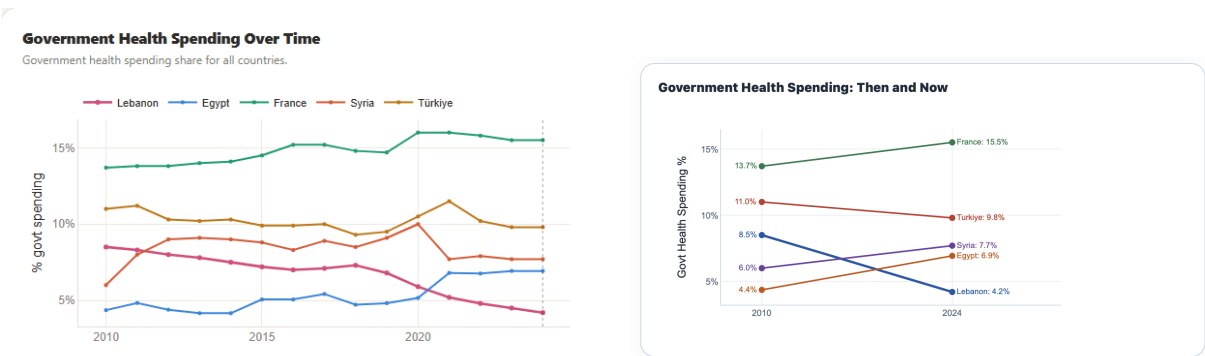


- (a) Before: scatter plot showing GDP per capita versus OOP burden. Exact comparison between countries requires estimating both x and y positions simultaneously.
- (b) After: horizontal bar chart with GDP capacity markers. OOP burden becomes easier to compare directly across countries while GDP capacity remains visible as a secondary reference marker.

Figure 35: Economic Capacity vs Household Health Burden redesign: replacing the scatter plot with a horizontal bar chart and reference markers to improve direct comparison of household burden levels.

Justification. The redesign improves comparison accuracy by using aligned bar lengths instead of two-dimensional point positions. According to effectiveness principles, aligned spatial position and length support more precise quantitative comparison than scattered point estimation across two axes. The GDP capacity markers preserve contextual economic information without reducing readability of the primary OOP burden comparison.

17.4 Change 3: Adding the Slope Chart for Government Health Spending

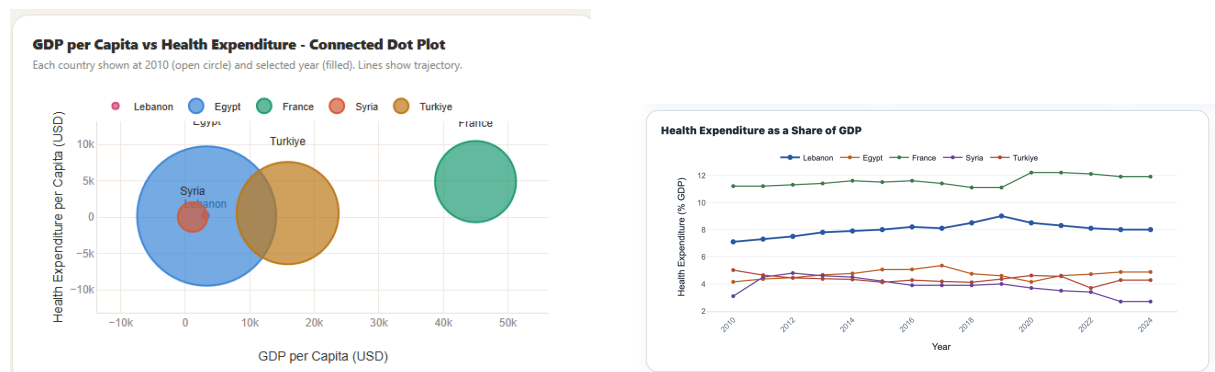


- (a) Before: multi-line time series showing government health spending trends for all countries between 2010 and 2024. Comparing overall change requires tracking multiple overlapping lines across time.
- (b) After: slope chart comparing only 2010 and 2024 values. Direction and magnitude of change become immediately visible for each country.

Figure 36: Government health spending redesign: replacing a multi-line temporal chart with a slope chart to improve direct comparison of change between two time points.

Justification. The redesign simplifies temporal comparison by reducing the amount of information users must visually track across the timeline. The slope chart emphasizes net change between the starting and ending years, making increases and decreases immediately interpretable while reducing clutter from overlapping time-series lines.

17.5 Change 4: Replacing the Connected Dot Plot



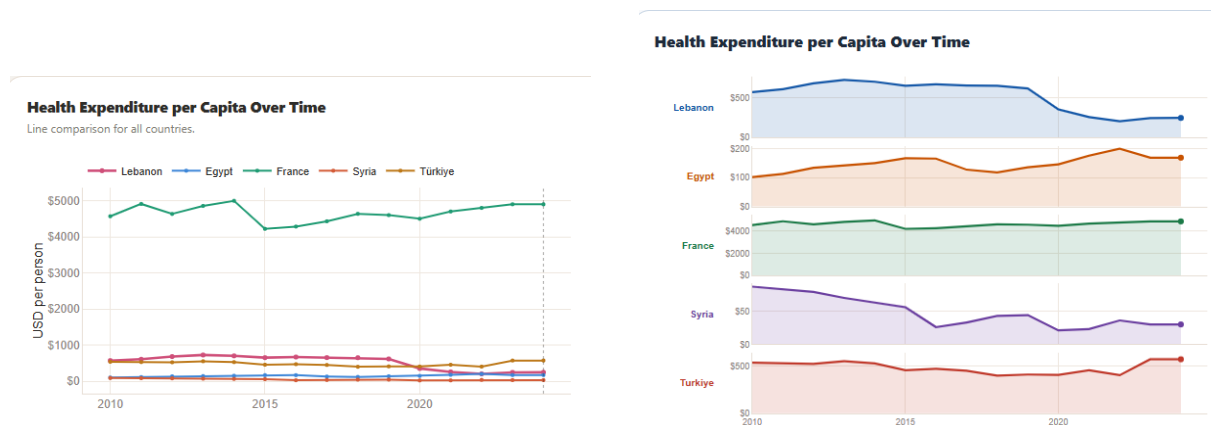
(a) Before: connected dot plot comparing GDP per capita and health expenditure per capita. Large bubble sizes and overlapping positions make comparison difficult.

(b) After: multi-line chart showing Health Expenditure as a Share of GDP over time. Trends and cross-country differences become easier to interpret.

Figure 37: Health expenditure redesign: replacing the connected dot plot with a temporal line chart focused on Health Expenditure as a Share of GDP.

Justification. The redesign improves readability by replacing overlapping bubble positions with aligned line trajectories across time. According to effectiveness principles, aligned spatial position supports more accurate comparison than bubble area and scattered point estimation. The new chart also introduces temporal continuity, allowing users to observe long-term healthcare spending trends and divergence between countries more clearly.

17.6 Change 5: Transitioning to Small Multiples for Comparative Trends



(a) Before: a single-axis line chart. France's significantly higher expenditure per capita compresses the data for other countries, making their individual trends nearly impossible to distinguish.

(b) After: faceted line charts (small multiples). Each country uses an independent y-axis scale, allowing for the clear visualization of internal fluctuations and local trends.

Figure 38: Health expenditure redesign: replacing a shared-scale line chart with small multiples to account for vast differences in spending magnitude.

Justification. The primary issue with the original chart was the high variation in data magnitude (France at \$5,000 vs. Syria at \$50), which resulted in “squashed” lines for four out of five countries. By implementing small multiples, we follow the principle of scale transparency. While the shared y-axis is removed, the aligned x-axes still permit temporal comparison, and the individual y-scales ensure that the specific “story” of each country such as Lebanon’s sharp decline after 2019 or Egypt’s steady growth is visually accessible. This eliminates the occlusion and scaling bias inherent in the previous version.

18 Discussion

The visual analysis presented in this report underscores the profound, multi-dimensional impact of Lebanon’s economic crisis on its healthcare system and demographic trajectory. By shifting from standard tabular reporting to an interactive visual dashboard, we enabled the immediate perception of critical anomalies, such as the hyperinflationary spike in the Medical CPI and the structural shift toward out-of-pocket health financing following the 2019 collapse.

The comparative framework also revealed important resilience factors and lagging indicators. Despite the catastrophic drop in GDP per capita and government health expenditure, Lebanon’s life expectancy has not immediately plummeted to match its lower-income regional peers, illustrating the complex temporal lag between economic shocks and mortality outcomes. This decoupling is itself an analytically significant non-obvious finding:

it suggests that the health consequences of the financial collapse may not yet be fully visible in aggregate statistics, and that the most severe impacts on life expectancy may still be forthcoming.

Several non-obvious patterns emerged from cross-tab exploration. First, Lebanon's fertility rate of 1.4 births/woman, when combined with a net migration rate of -12.5 per 1,000, implies a structural population contraction that is not yet visible in absolute population figures due to the large base. If these trends continue, Lebanon's healthcare system will face declining revenues and a shrinking working-age population precisely when the chronic disease burden from aging increases. Second, Lebanon's Health Affordability Index (7.8%) is higher than Egypt's (5.1%), meaning that Lebanese households devote a larger share of national income to healthcare than Egyptians do, even though Lebanon's OOP burden has worsened significantly post-crisis. This suggests that the healthcare sector has absorbed a disproportionate share of household expenditure during the economic collapse.

Based on these findings, three evidence-based recommendations follow. First, government health spending as a share of government expenditure must be restored from the current 4.2% toward the 2010 baseline of 8.5%, as the current level has pushed OOP burden to 35.7% and risks making healthcare unaffordable for lower-income households. Second, hospital infrastructure policy should address the geographic concentration identified in Chart 11: Mount Lebanon and Beirut together account for 90 of the 171 hospitals, while Nabatieh and Bekaa are significantly underserved relative to their populations. Third, emigration of healthcare workers, visible in the net migration data, requires targeted retention programs, as the loss of trained medical staff compounds the resource scarcity caused by the fiscal retreat from healthcare funding.

The iterative redesign process proved essential. The transition to small multiples for health expenditure and year-over-year bar charts for inflation resolved extreme disparities in data magnitude between European benchmarks and crisis-affected nations, adhering strictly to principles of expressiveness and effectiveness.

19 Conclusion

The Lebanon Healthcare & Population Impact Dashboard successfully fulfills its primary objective: providing a comprehensive, exploratory tool for stakeholders to analyze a complex public health crisis. By applying principled data visualization techniques, including intuitive color mappings, redundant encodings, and coordinated global interactions, the dashboard transforms raw, disconnected datasets into a cohesive narrative.

Future iterations of this project could enhance its analytical depth by incorporating more

granular sub-national data, such as district-level morbidity rates or population-normalized hospital rates by governorate, which would justify the addition of a choropleth map layer. Integrating predictive modeling to forecast the long-term impacts of the current emigration and fertility trends would extend the dashboard from exploratory analysis into decision support. Ultimately, this visualization system stands as a vital mechanism for understanding the ongoing Lebanese crisis through a comparative, data-driven lens.

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